

Deep Learning for Data Science

DS 542

<https://dl4ds.github.io/fa2025/>

Supervised Learning

Supervised learning

- Examples
- Terminology
- Notation
 - Model
 - Loss function
 - Training
 - Testing
- 1D Linear regression example
 - Model
 - Loss function
 - Training
 - Testing

Artificial intelligence

Machine learning

Supervised
learning

Unsupervised
learning

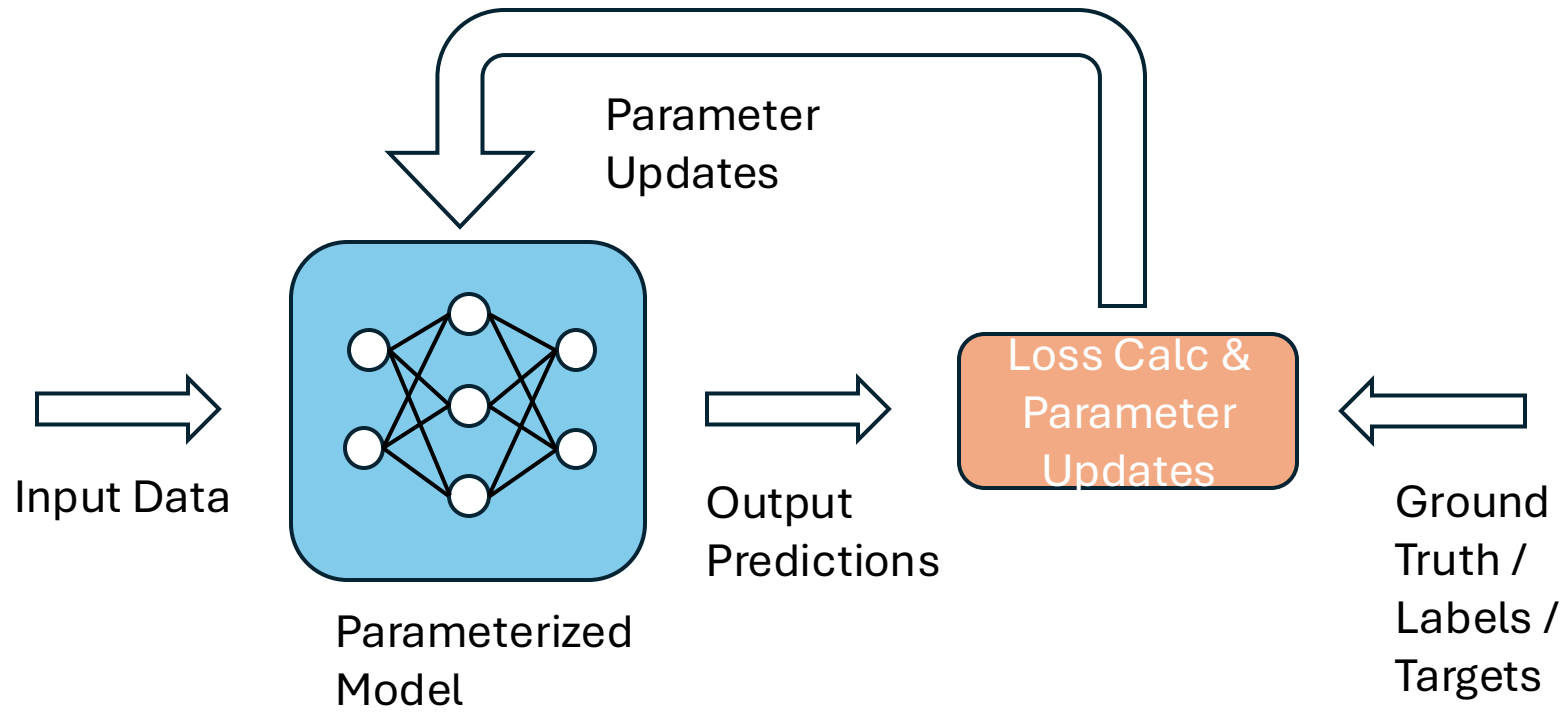
Reinforcement
learning

Deep learning

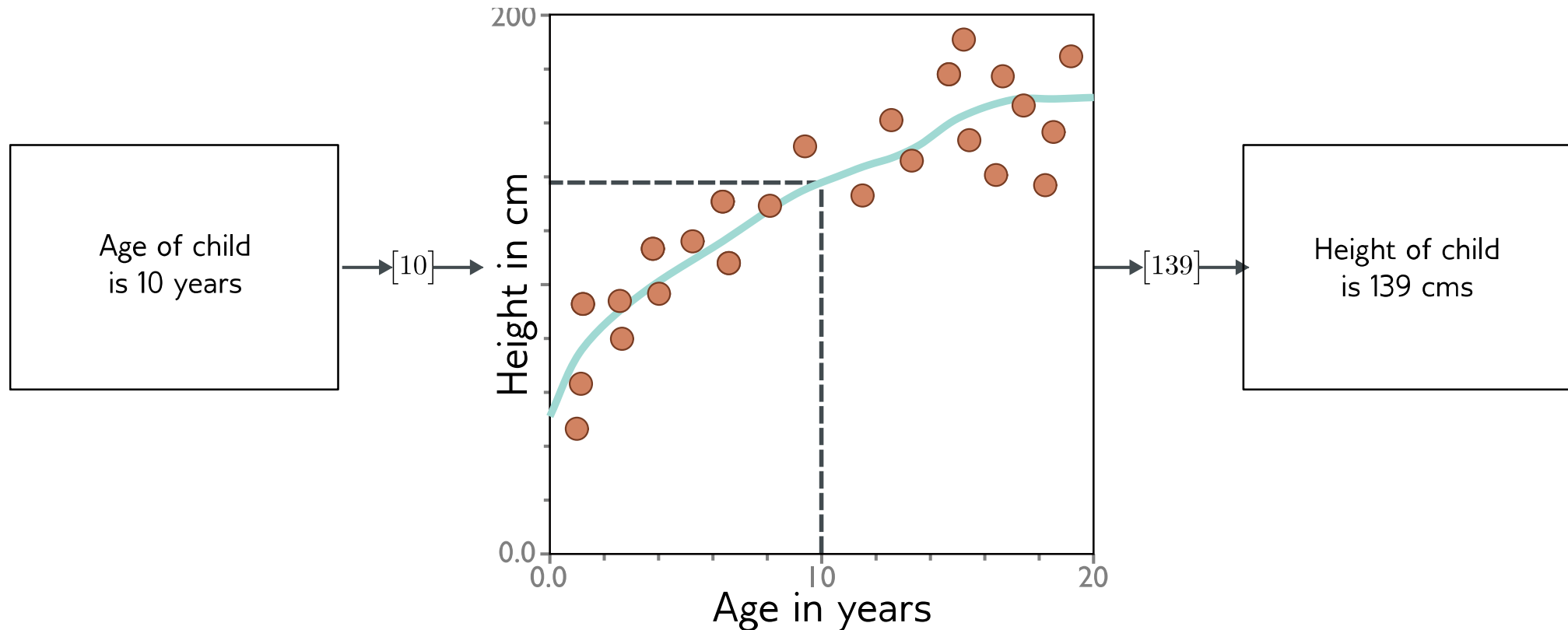


Supervised learning

- Define a mapping from input to output
- Learn this mapping from paired input/output data examples

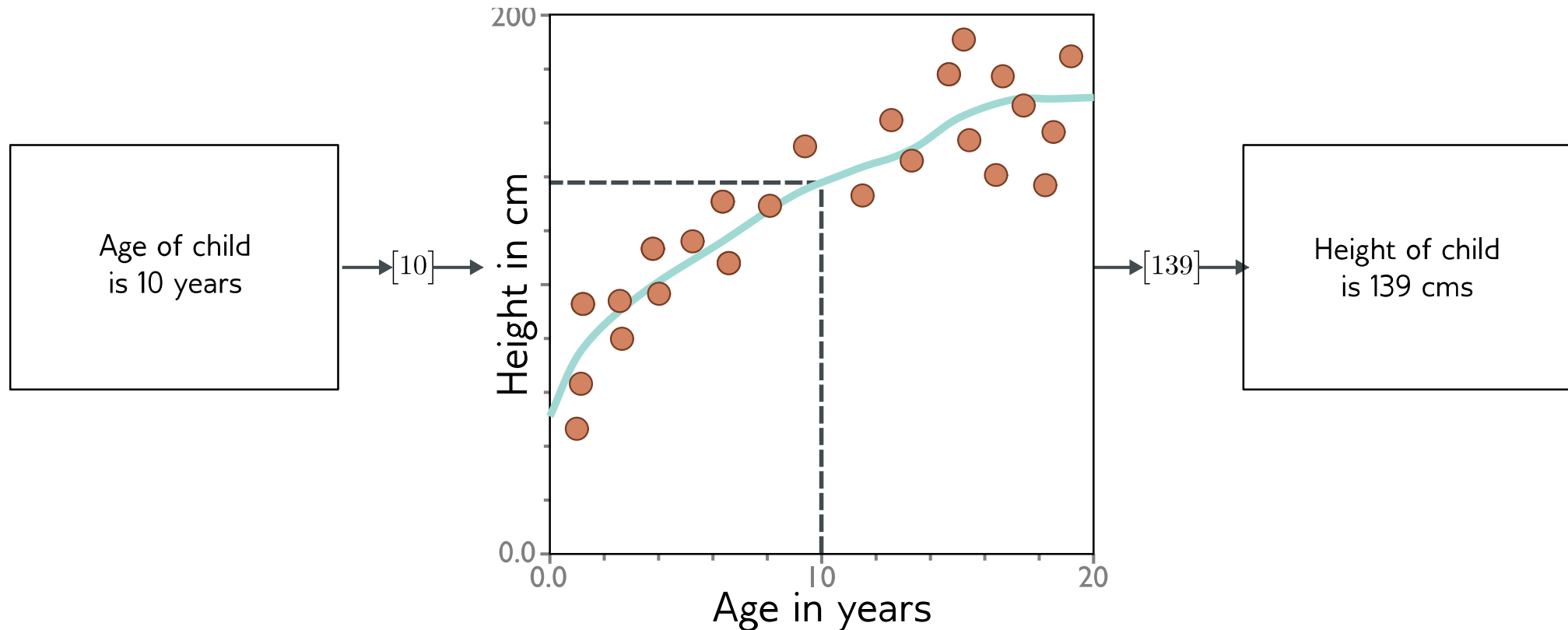


What is a supervised learning model?



- An equation relating input (age) to output (height)
- Search through family of possible equations to find one that fits training data well

What is a supervised learning model?



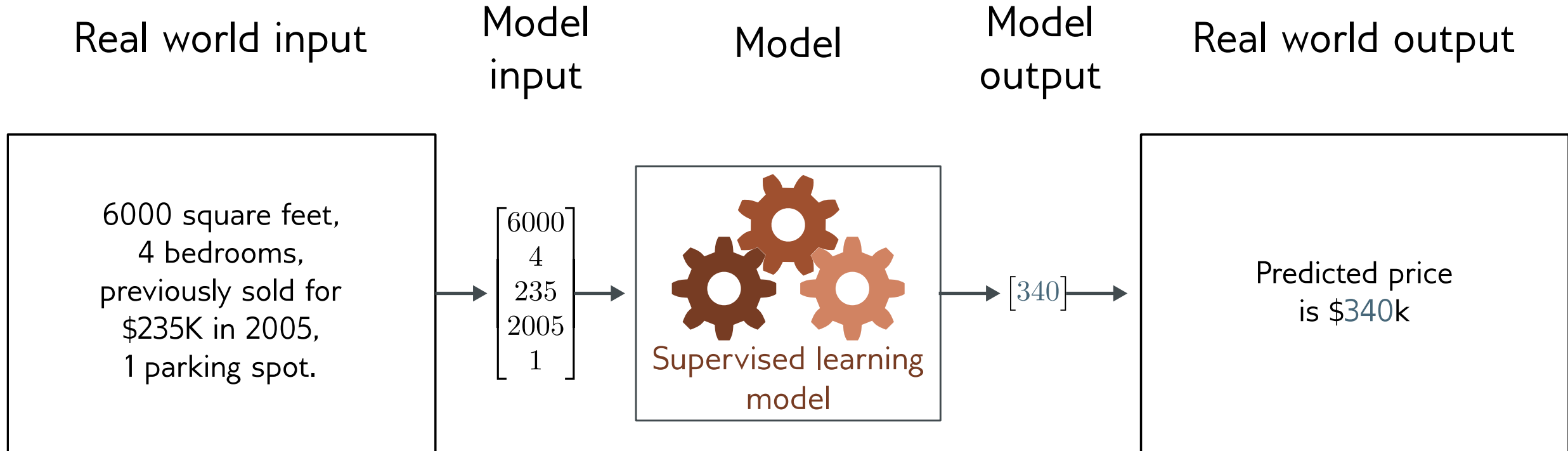
- Deep neural networks are just a very flexible family of equations
- Fitting deep neural networks = “Deep Learning”

Prediction Types

- Regression
 - Prediction a continuous valued output
- Classification
 - Assigning input to one of a finite number of classes or categories
 - Two classes are a special case

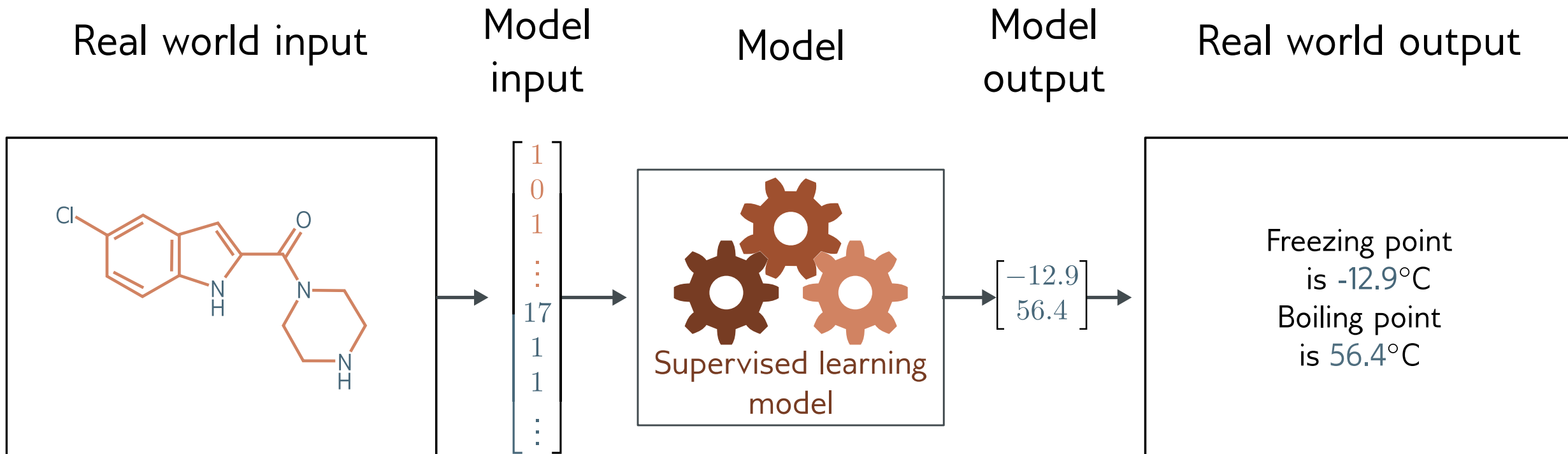
Can be univariate (one output) or multivariate (more than one output)

Regression



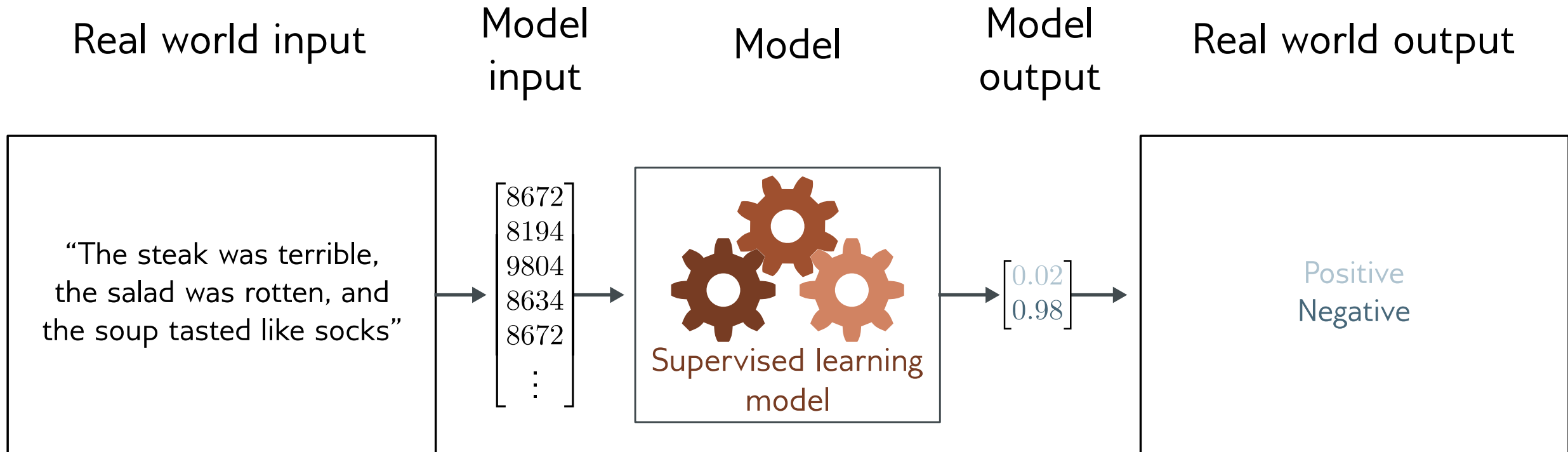
- Univariate regression problem (one output, real value)
- Fully connected network

Graph regression



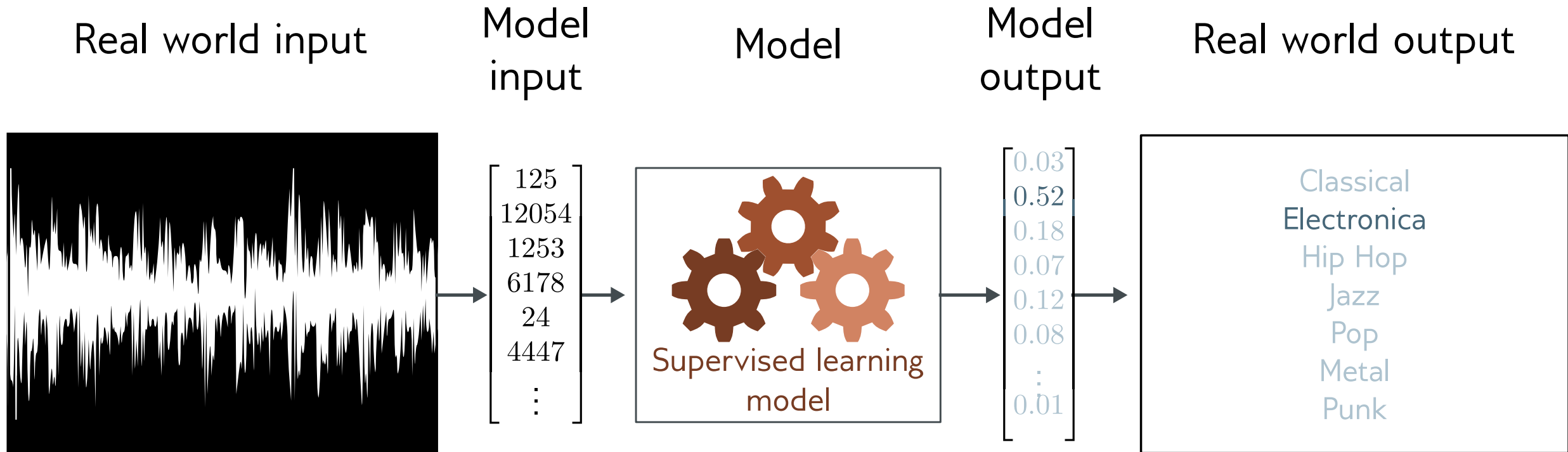
- Multivariate regression problem (>1 output, real value)
- Graph neural network

Text classification



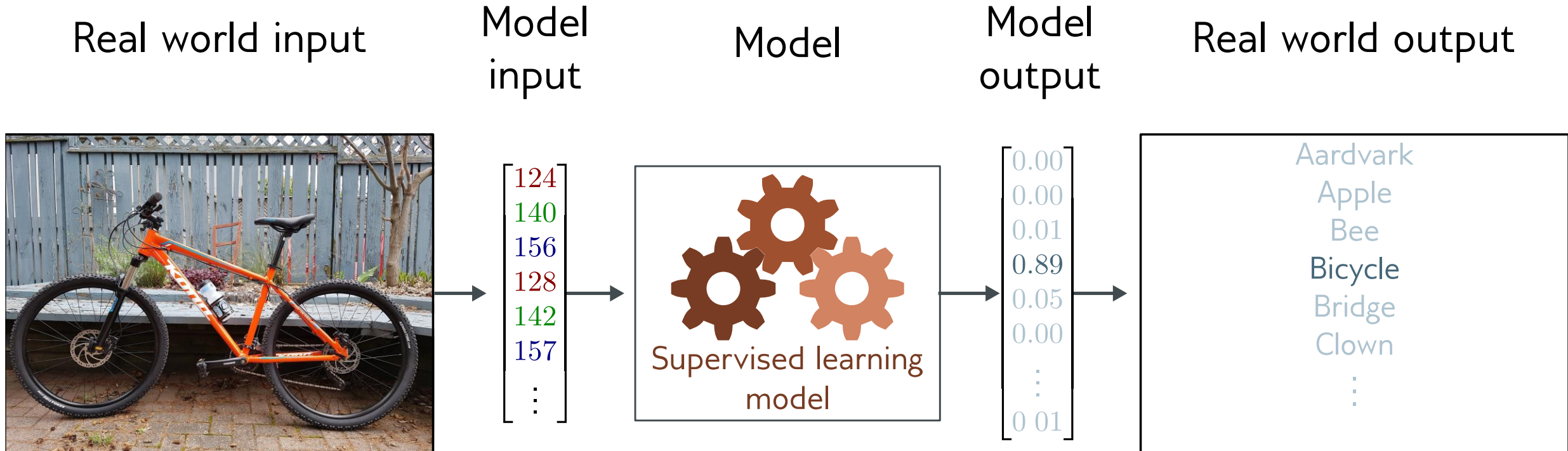
- Binary classification problem (two discrete classes)
- Transformer network

Music genre classification



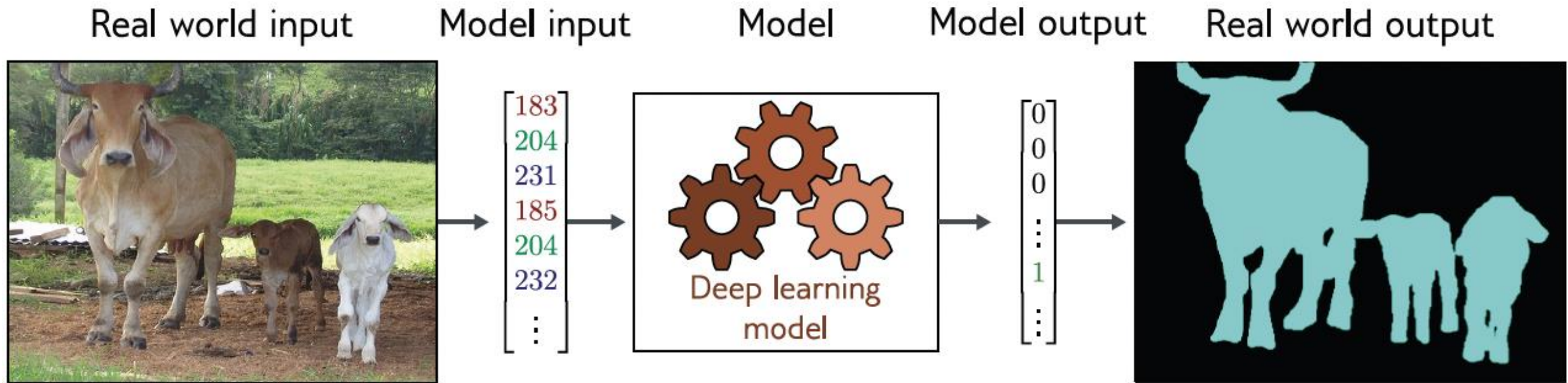
- Multiclass classification problem (discrete classes, >2 possible values)
- Recurrent neural network (RNN)

Image classification



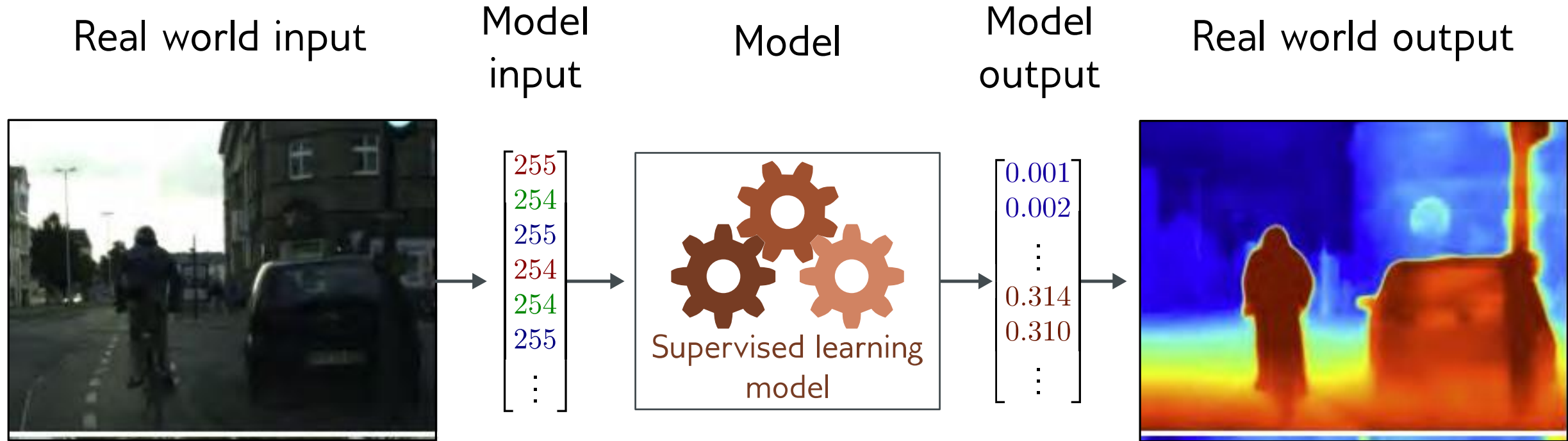
- Multiclass classification problem (discrete classes, >2 possible classes)
- Convolutional network

Image segmentation



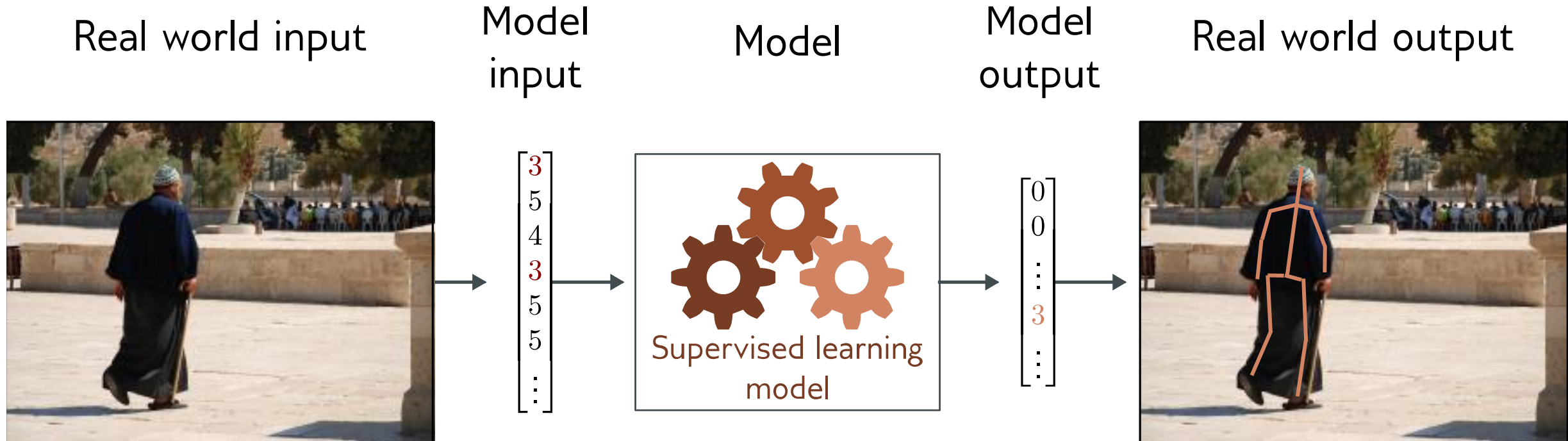
- Multivariate binary classification problem (many outputs, two discrete classes)
- Convolutional encoder-decoder network

Depth estimation



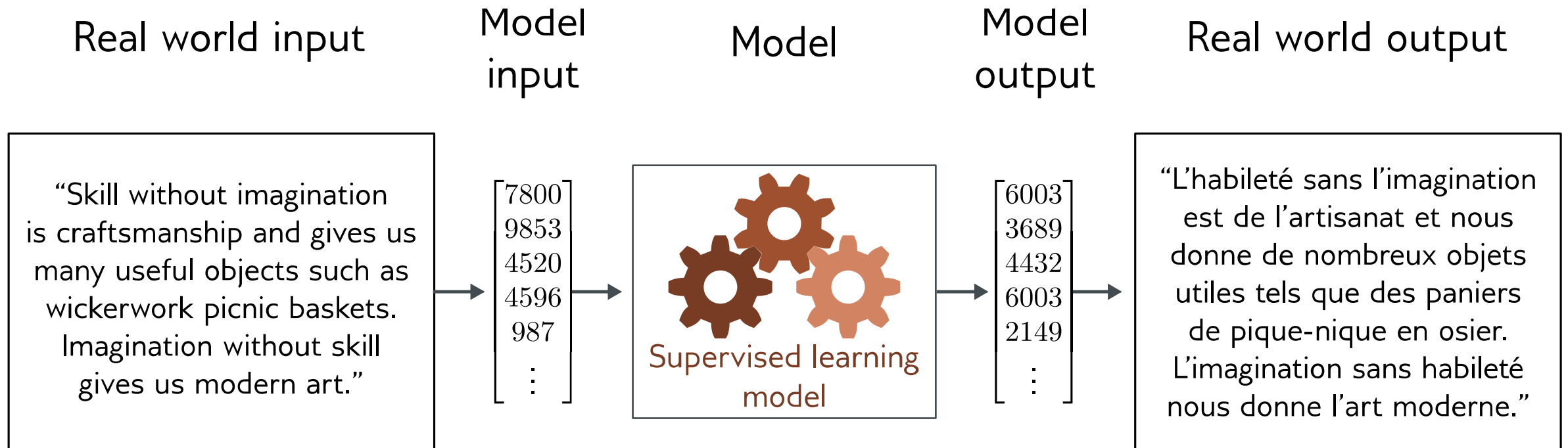
- Multivariate regression problem (many outputs, continuous)
- Convolutional encoder-decoder network

Pose estimation



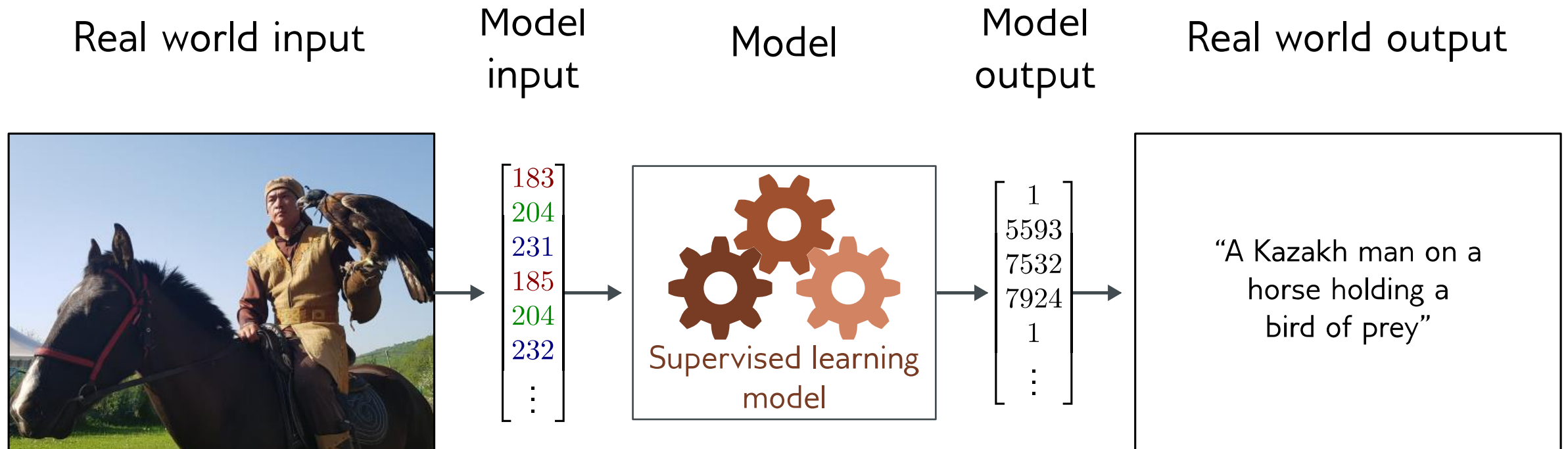
- Multivariate regression problem (many outputs, continuous)
- Convolutional encoder-decoder network

Translation



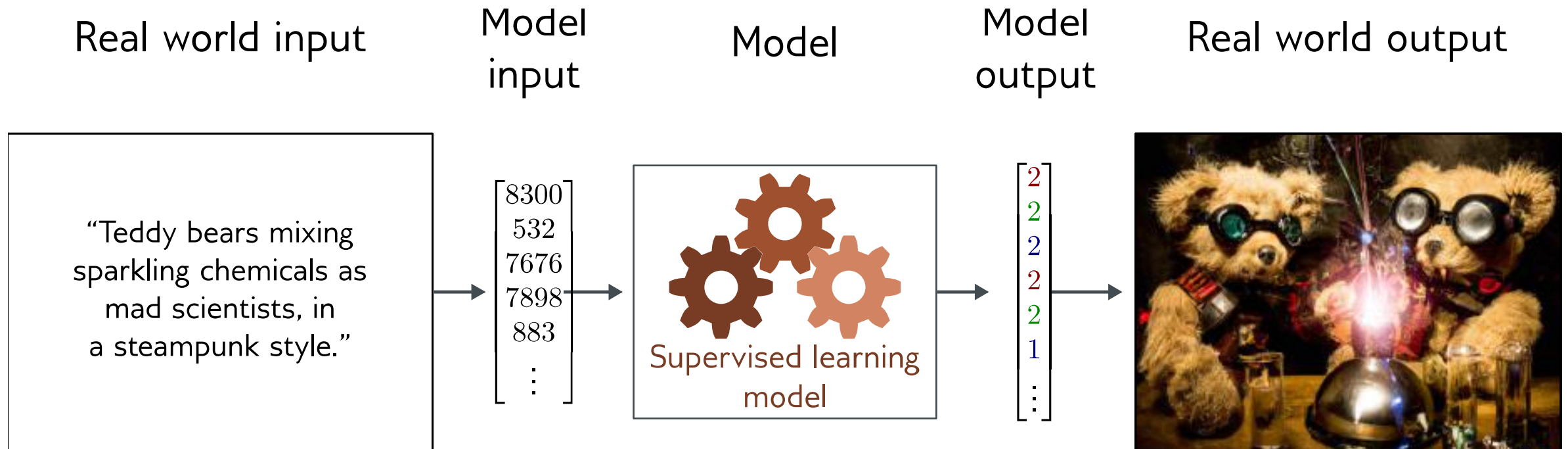
- Encoder-Decoder Transformer Networks

Image captioning

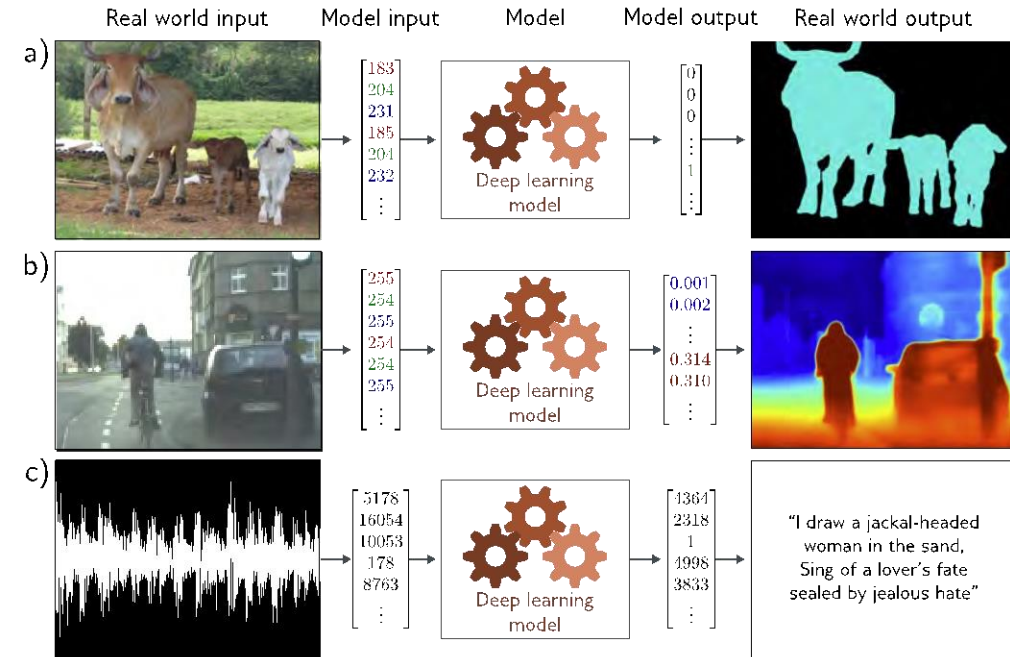
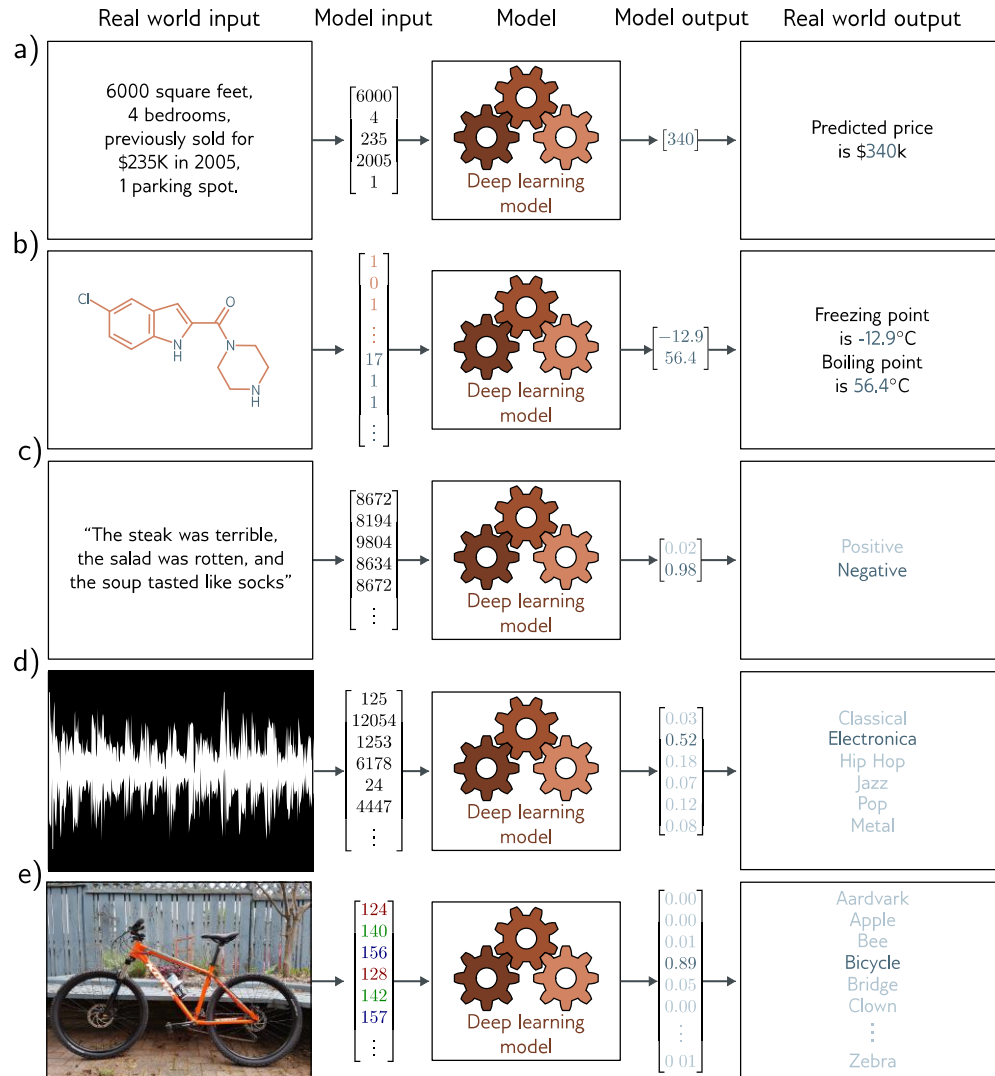


- E.g. CNN-RNN, LSTM, Transformers

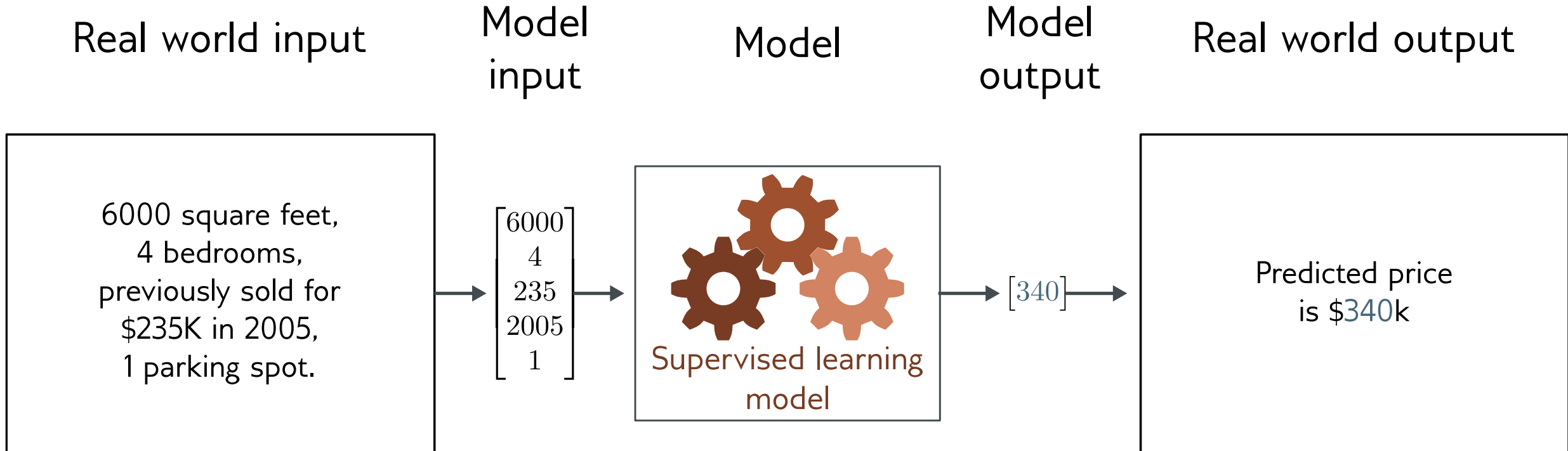
Image generation from text



Supervised Learning Classification and Regression Applications



Regression



- Univariate regression problem (one output, real value)

Any Questions?

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Supervised learning terminology

- **Supervised learning model** = mapping from one or more inputs to one or more outputs
- Model is a family of equations → “**inductive bias**”
- Computing the outputs from the inputs → **inference**
- Model also includes **parameters**
- Parameters affect outcome of equation
- **Training** a model = finding parameters that predict outputs “well” from inputs for **training** and **evaluation datasets** of input/output pairs

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Notation:

- Input:

x



Variables always Roman letters

- Output:

y

Normal lower case = scalar
Bold lower case = vector
Capital Bold = matrix

- Model:

y = **f**[**x**]



Functions always square brackets

Normal lower case = returns scalar
Bold lower case = returns vector
Capital Bold = returns matrix²⁵

Notation example:

- Input:

$$\mathbf{x} = \begin{bmatrix} \text{age} \\ \text{mileage} \end{bmatrix}$$



Vector:
Structured or
tabular data

- Output:

$$y = [\text{price}]$$



Scalar output

- Model:

$$y = f[\mathbf{x}]$$



Scalar output
function
(with vector input)

Model

- Parameters:

ϕ

Parameters always
Greek letters

- Model :

$$y = f[x, \phi]$$

Data Set and Loss function

- Training dataset of I pairs of input/output examples:

$$\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I$$

Data Set and Loss function

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- **Loss function** or **cost function** measures how bad model is:

$$L \left[\underbrace{\phi, f[\mathbf{x}, \phi]}_{\text{model}}, \underbrace{\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I}_{\text{train data}} \right]$$

Data Set and Loss function

- Training dataset of I pairs of input/output examples:

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$$L \left[\underbrace{\phi, f[\mathbf{x}, \phi]}_{\text{model}}, \underbrace{\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I}_{\text{train data}} \right]$$

or for short:

$$L[\phi]$$

← Returns a scalar that is smaller when model maps inputs to outputs better

Training

- Loss function:

$$L[\phi]$$


Returns a scalar that is smaller when model maps inputs to outputs better

- Find the parameters that minimize the loss:

$$\hat{\phi} = \operatorname{argmin}_{\phi} [L[\phi]]$$

Any Questions?

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Example: 1D Linear regression model

- Model:

$$\begin{aligned}y &= f[x, \phi] \\ &= \phi_0 + \phi_1 x\end{aligned}$$

- Parameters

$$\phi = \begin{bmatrix} \phi_0 \\ \phi_1 \end{bmatrix}$$

← y-offset

← slope

Example: 1D Linear regression model

- Model:

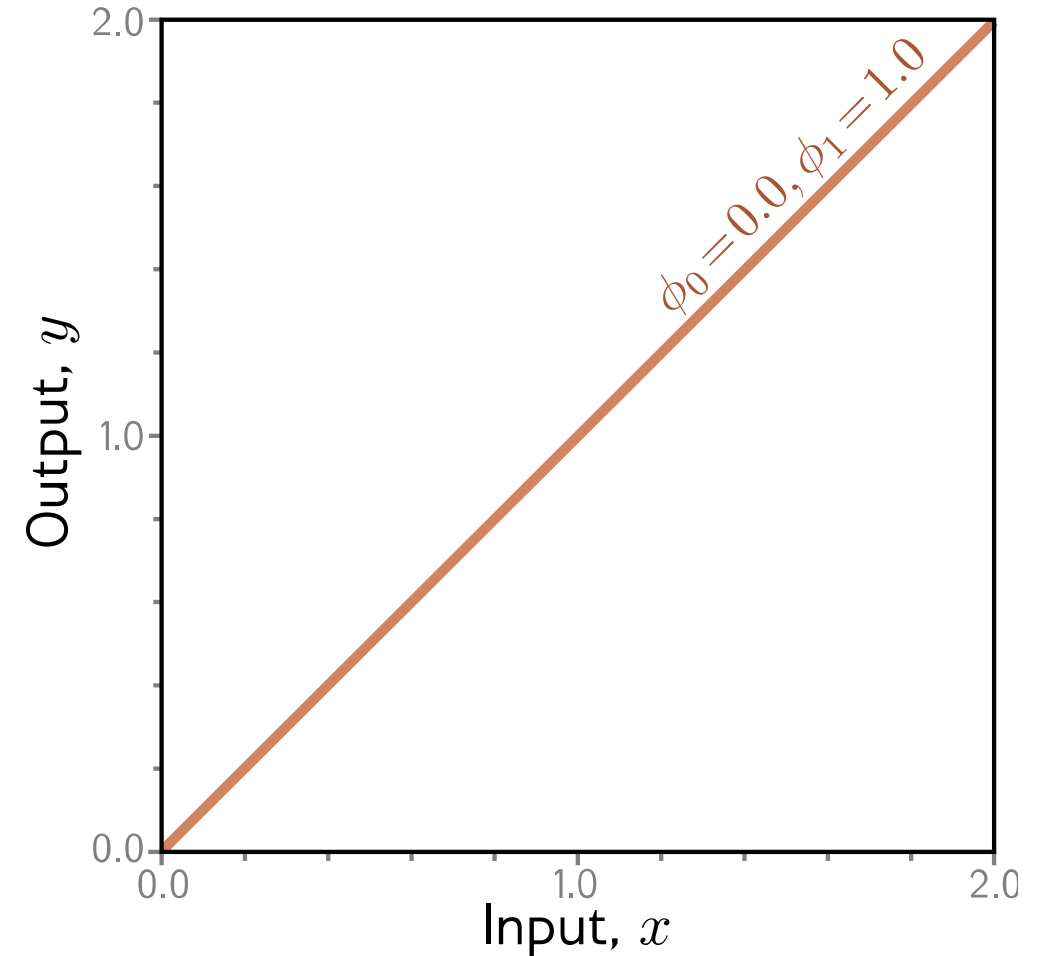
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Example: 1D Linear regression model

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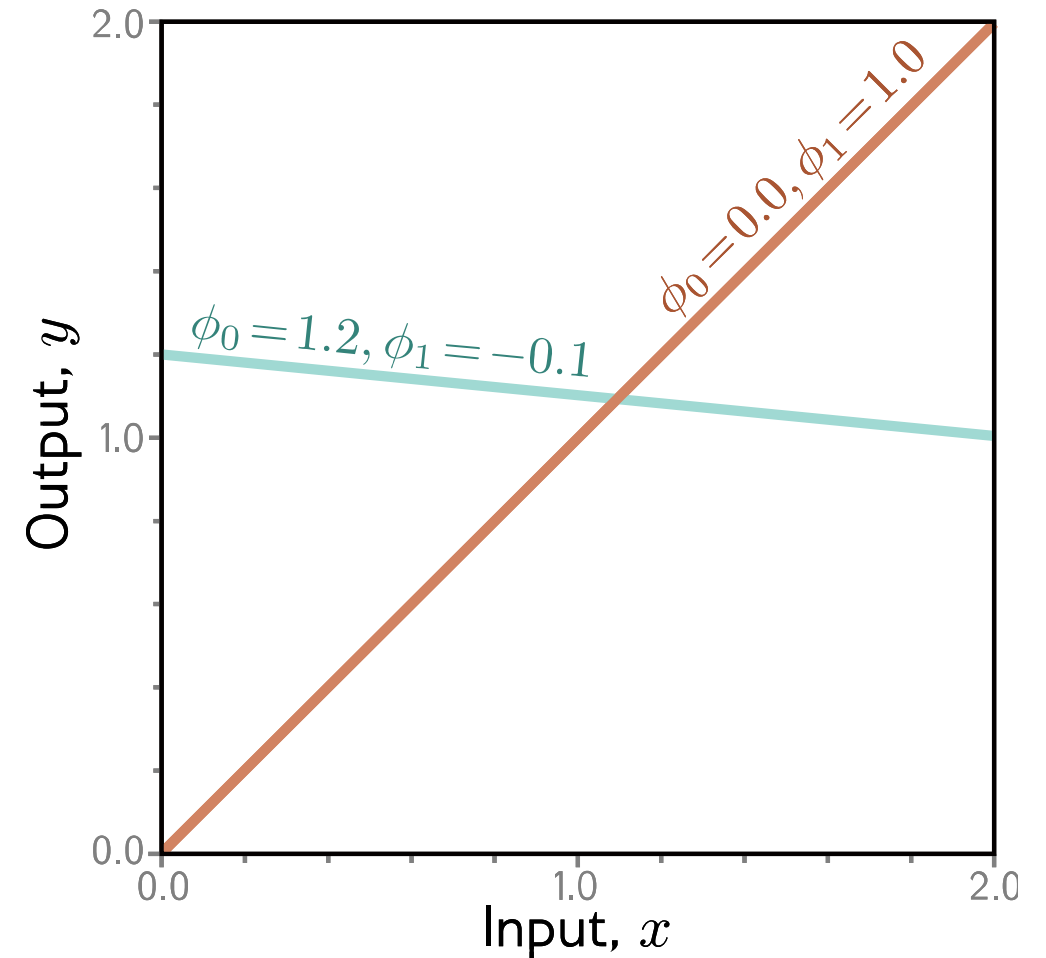
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Example: 1D Linear regression model

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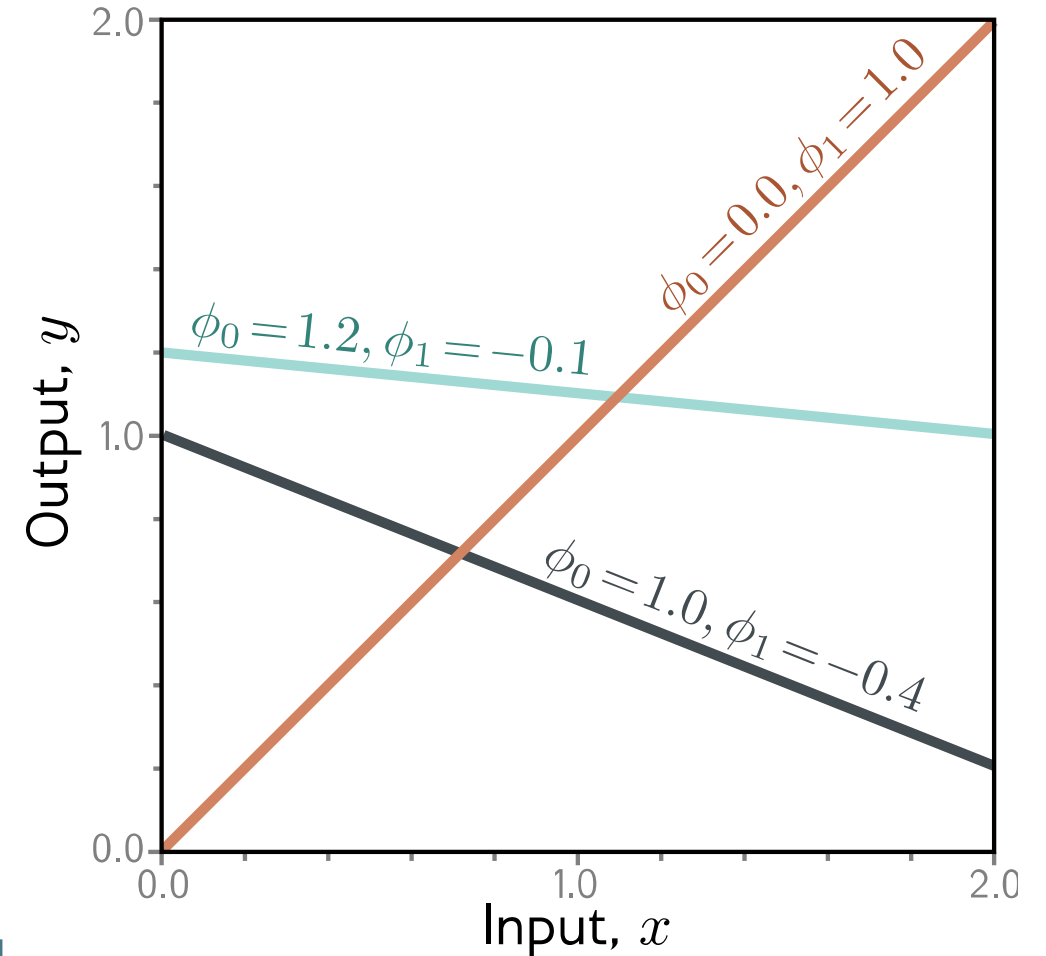
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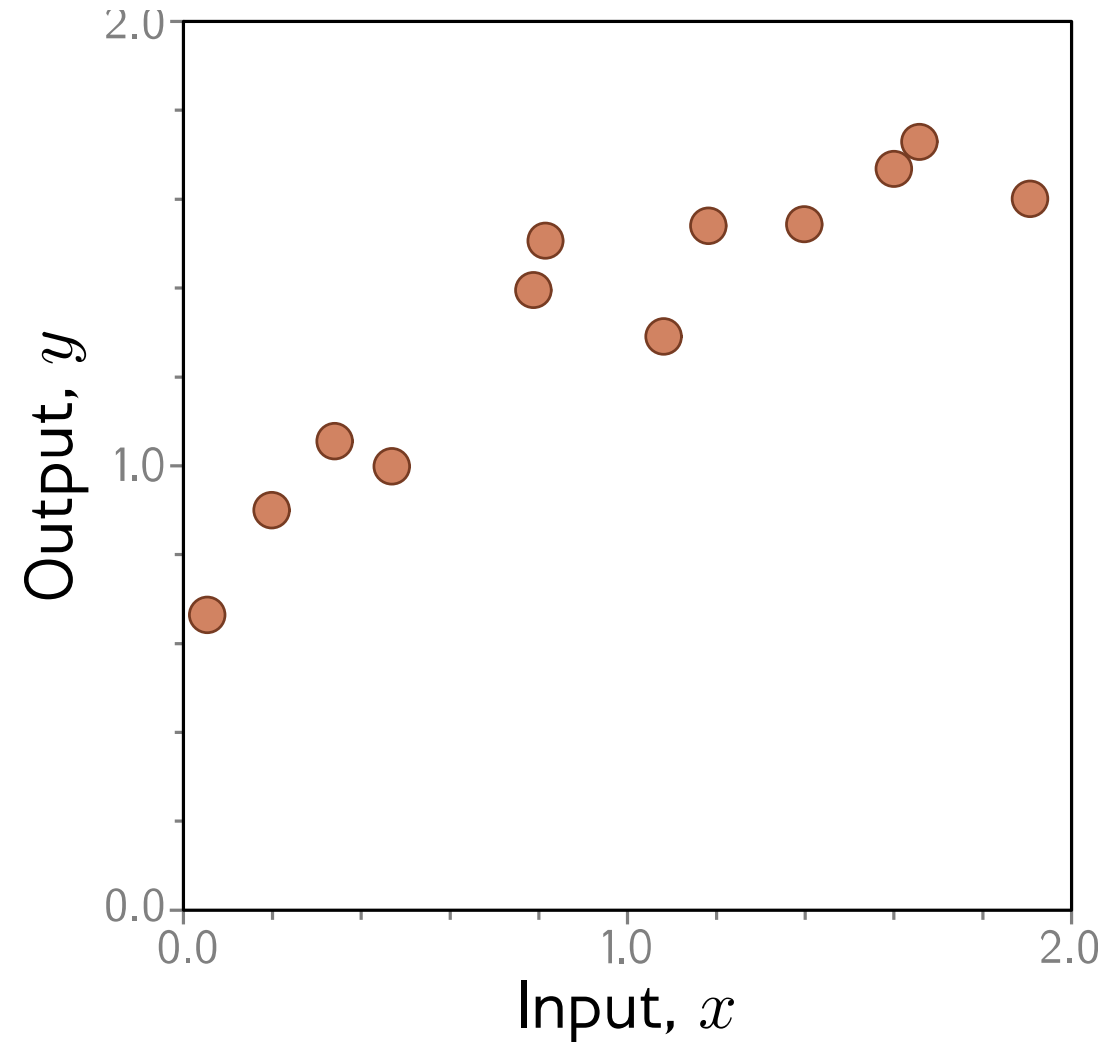
← y-offset

← slope

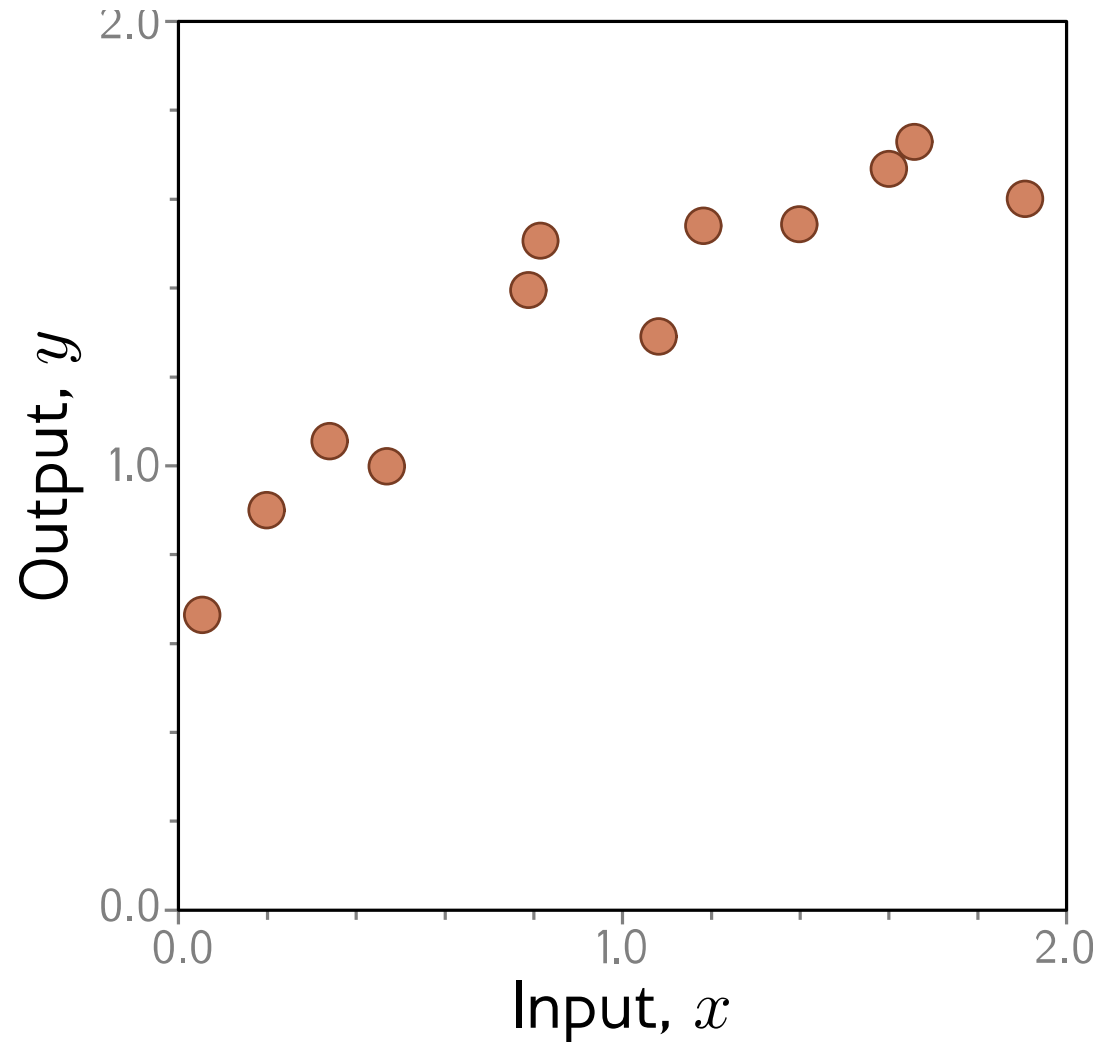


[Interactive Figure 2.1](#)

Example: 1D Linear regression training data



Example: 1D Linear regression training data

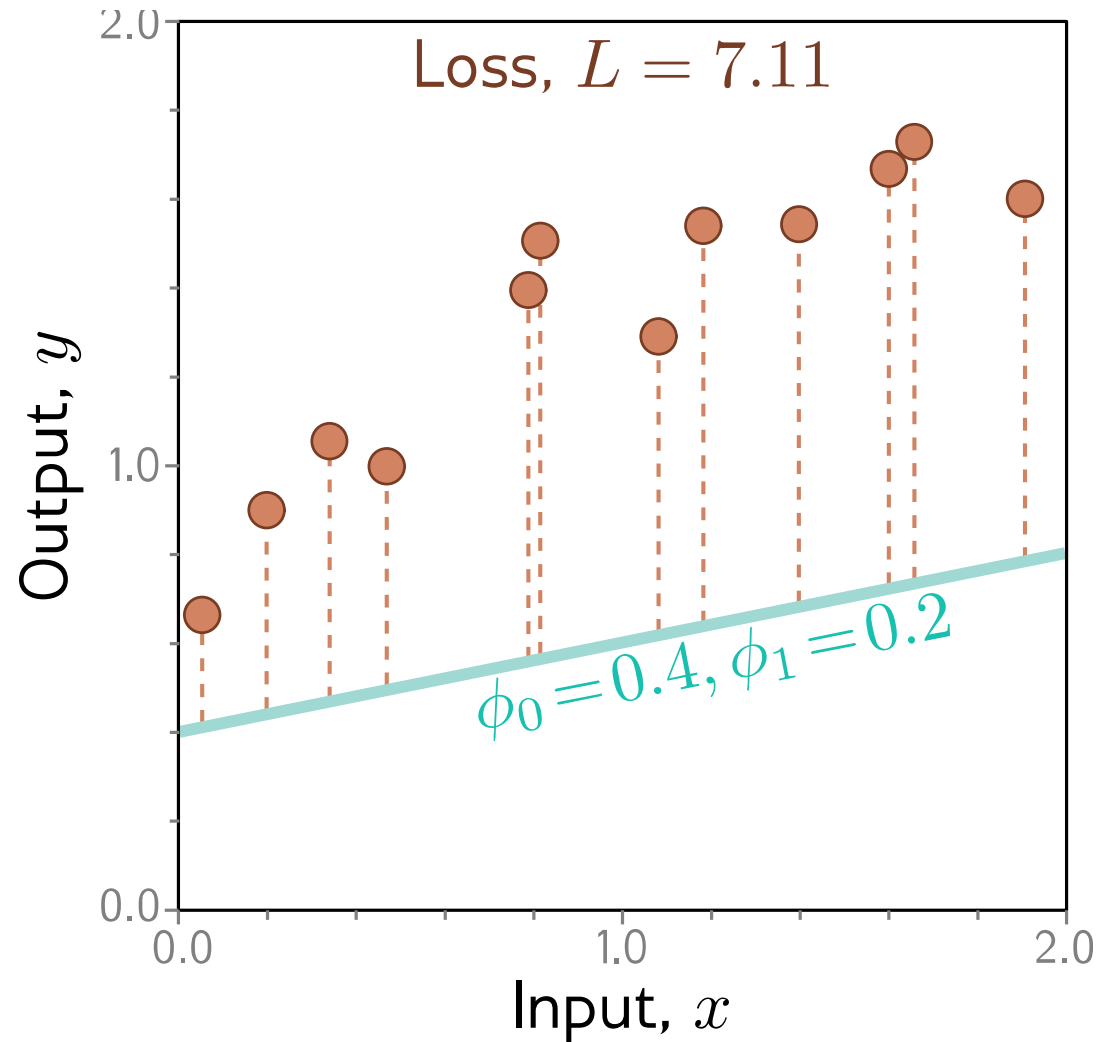


Loss function:

$$L[\phi] = \sum_{i=1}^I (f[x_i, \phi] - y_i)^2$$
$$= \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i)^2$$

“Least squares loss function”

Example: 1D Linear regression loss function

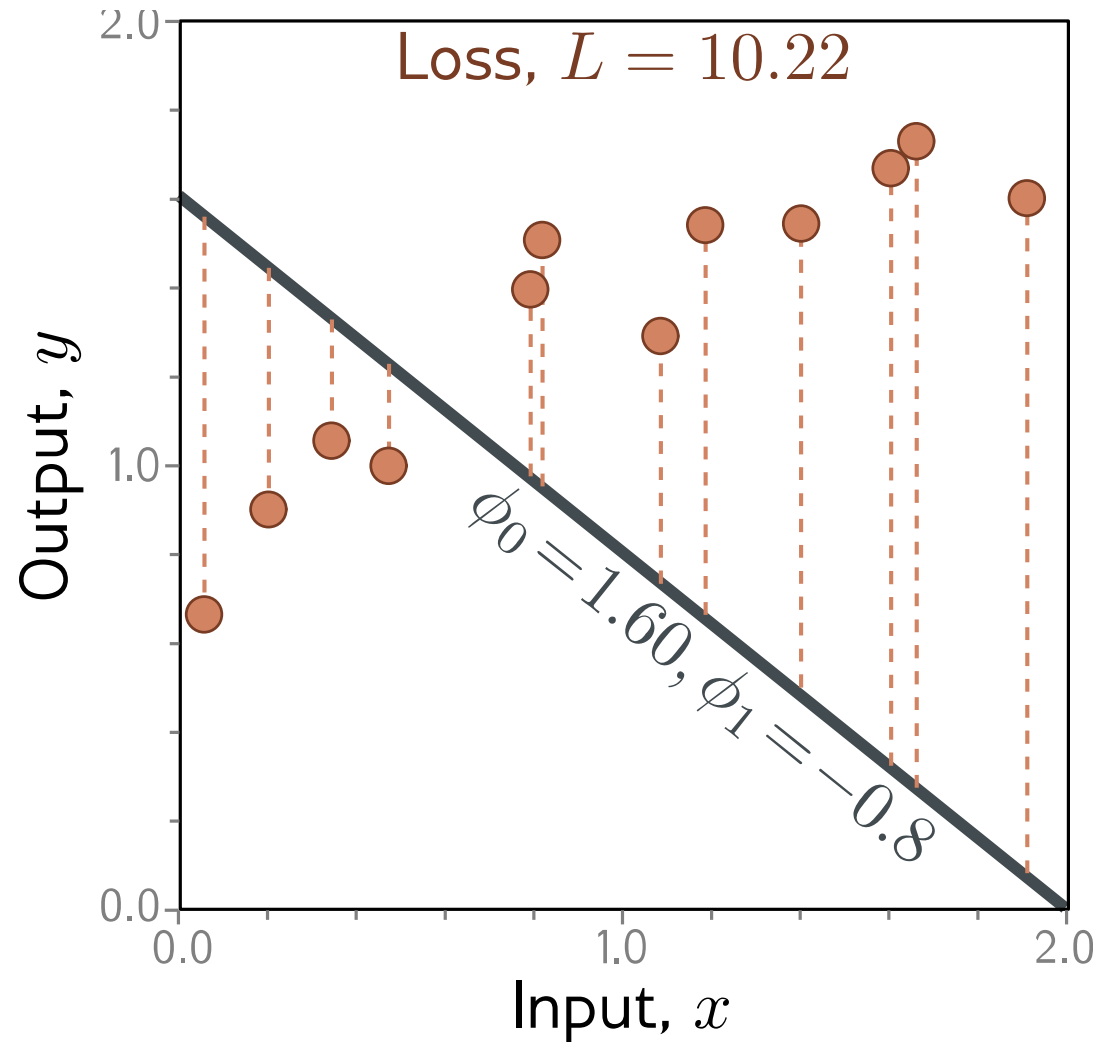


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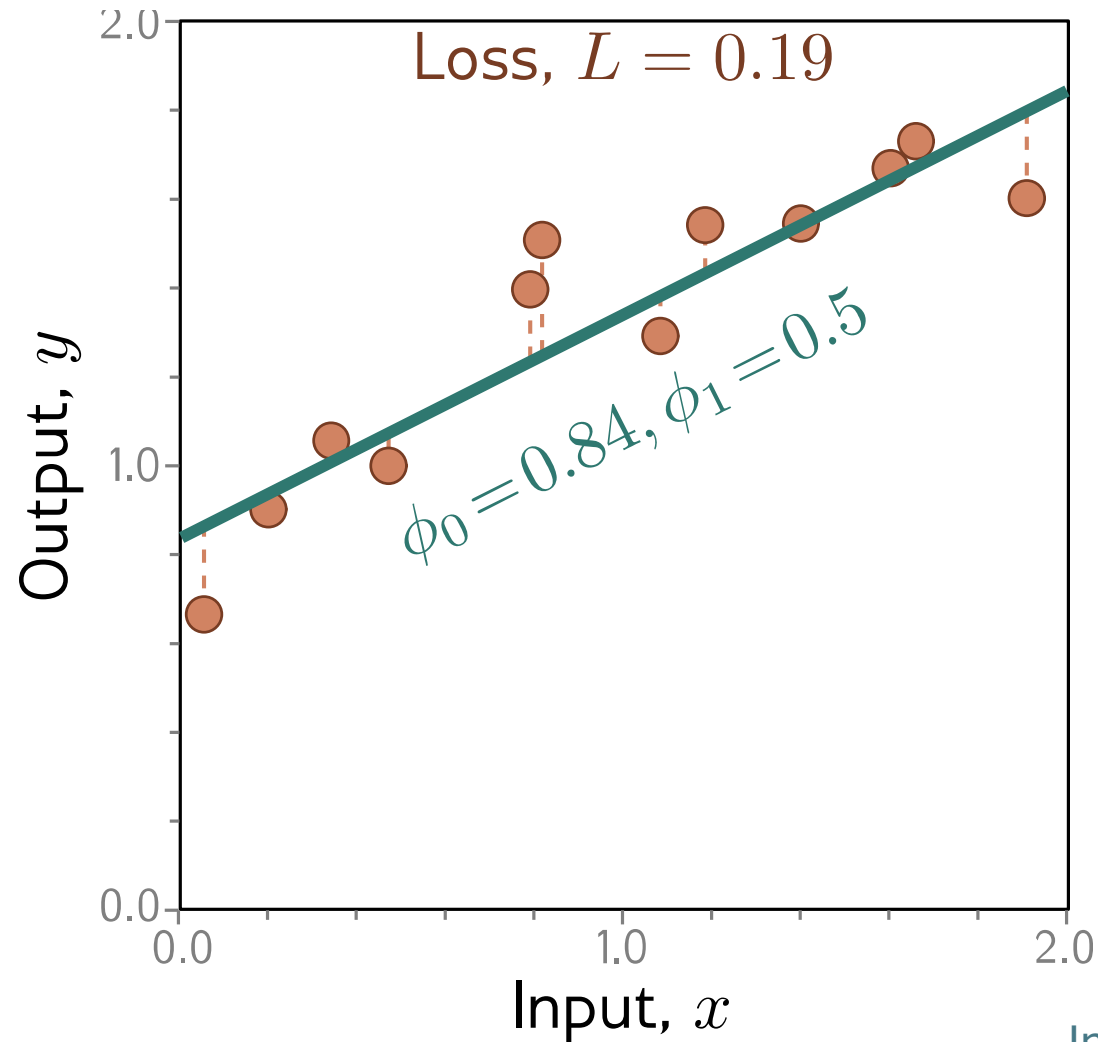


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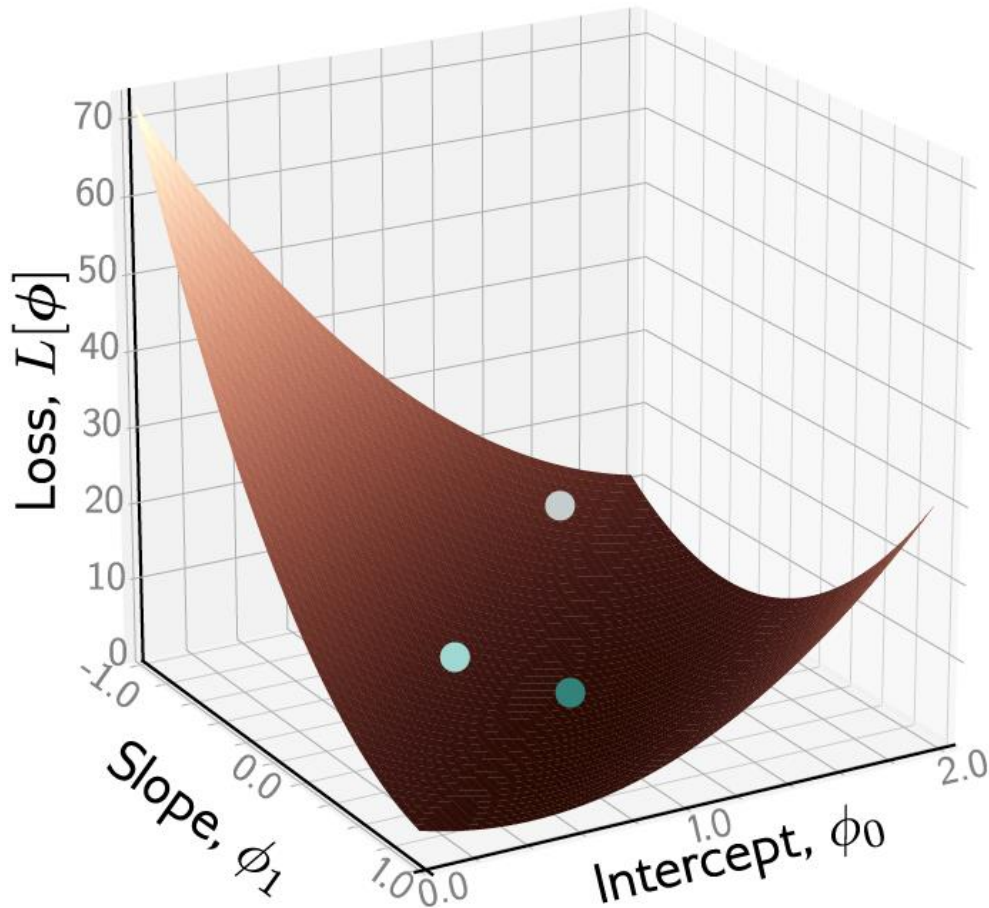
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“Least squares loss
function”

[Interactive Figure 2.2](#)

Example: 1D Linear regression loss function

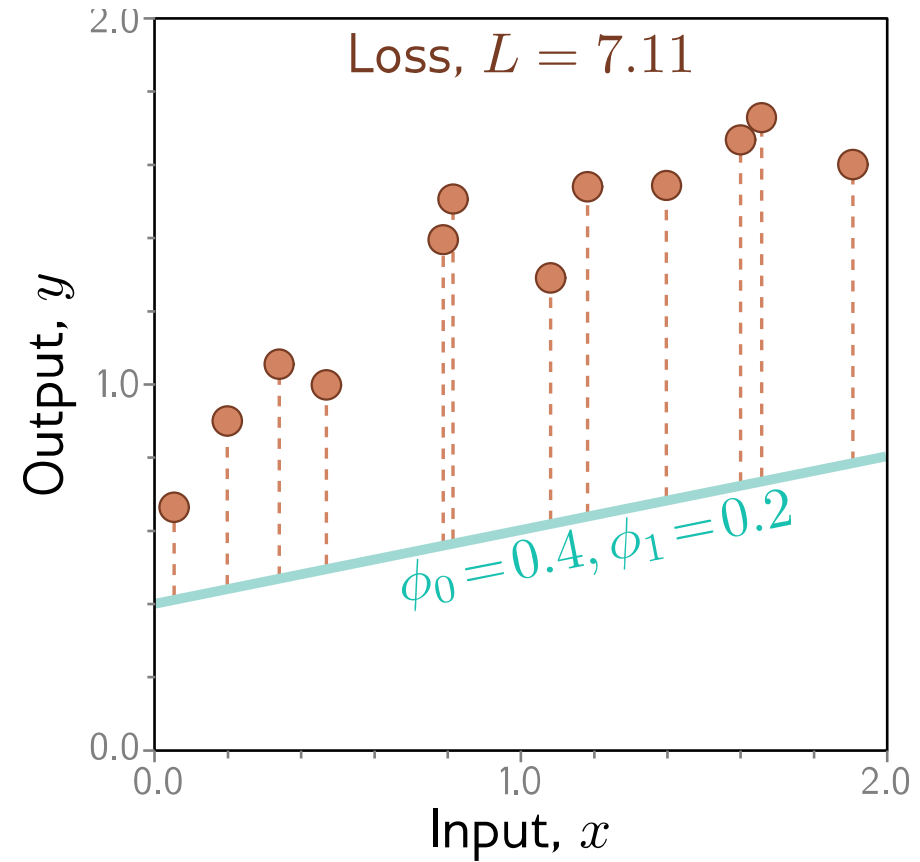
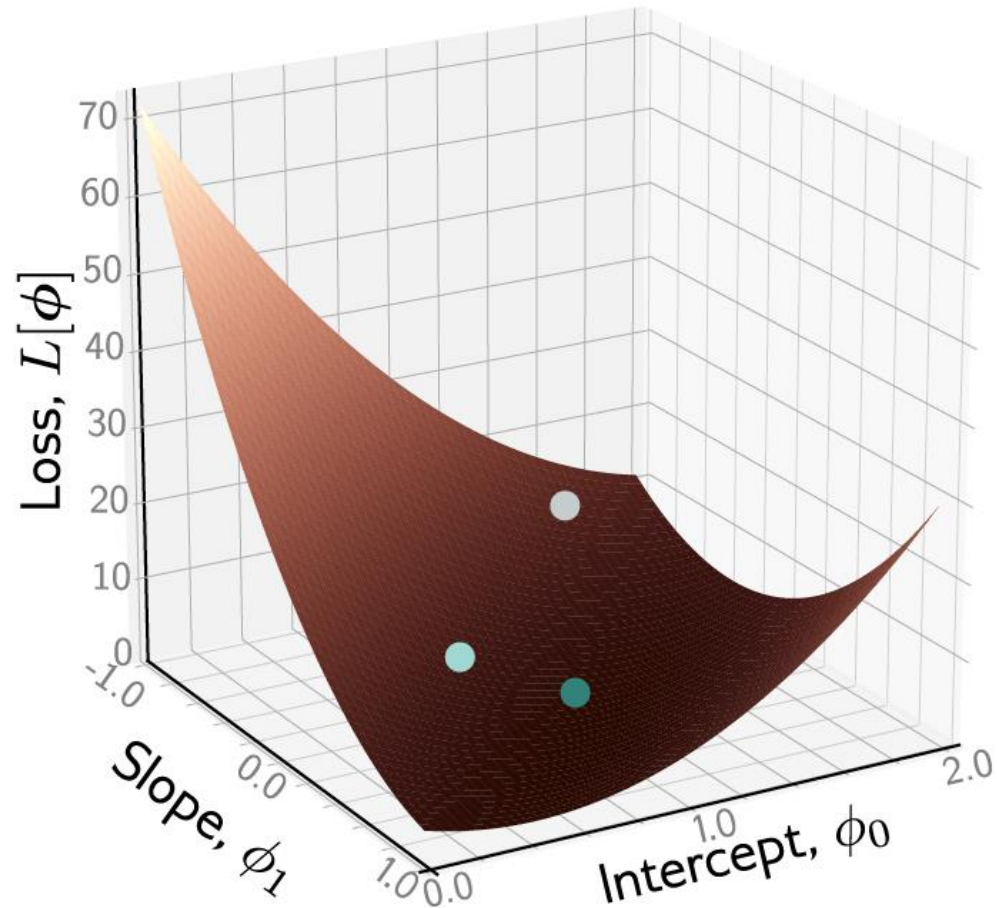


Loss function:

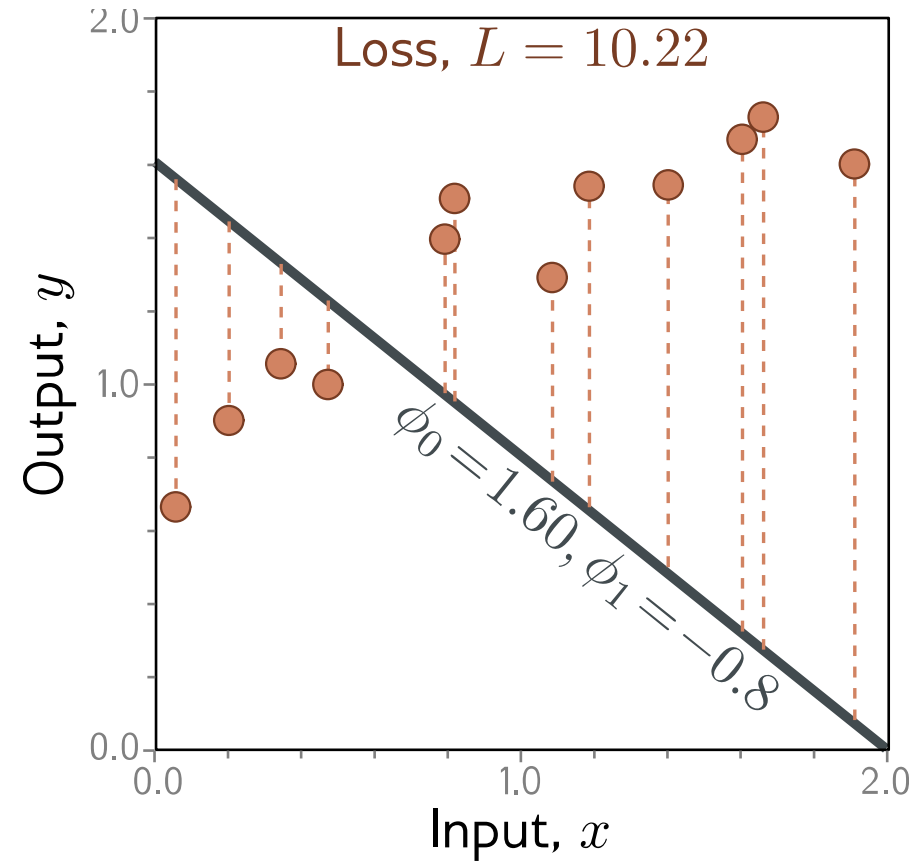
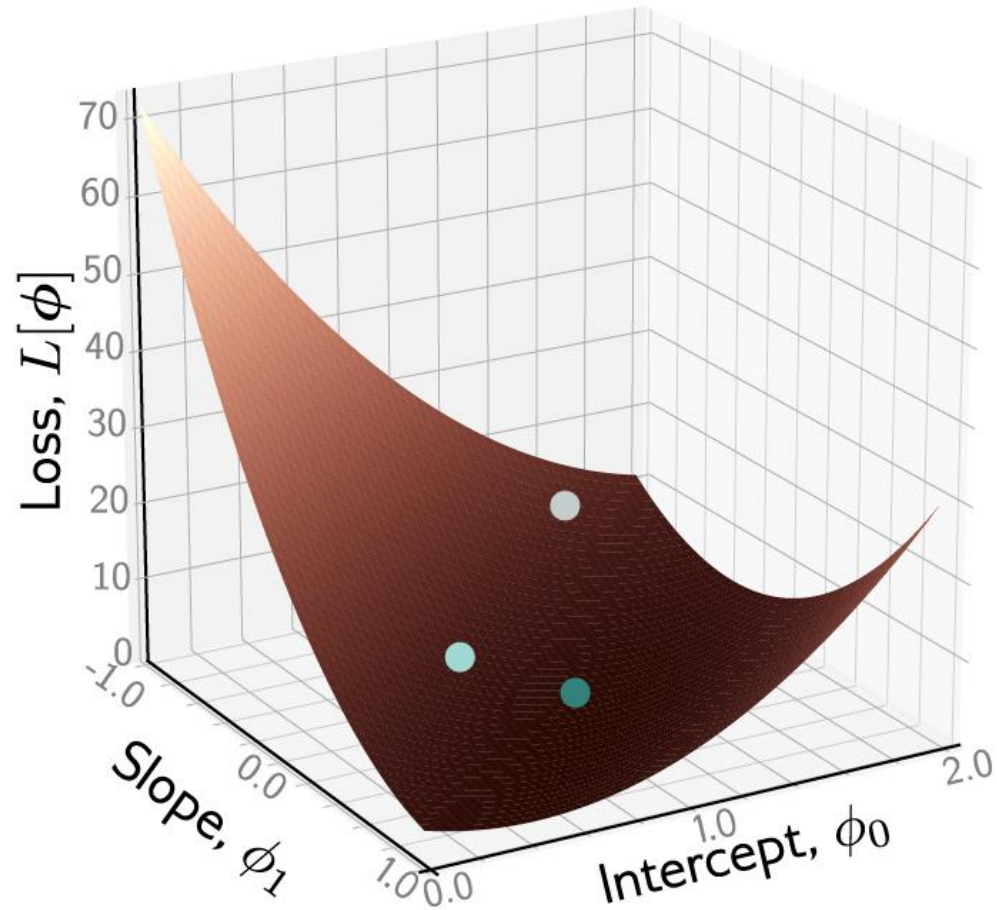
$$\begin{aligned} L[\phi] &= \sum_{i=1}^I (f[x_i, \phi] - y_i)^2 \\ &= \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i)^2 \end{aligned}$$

“Least squares loss
function”

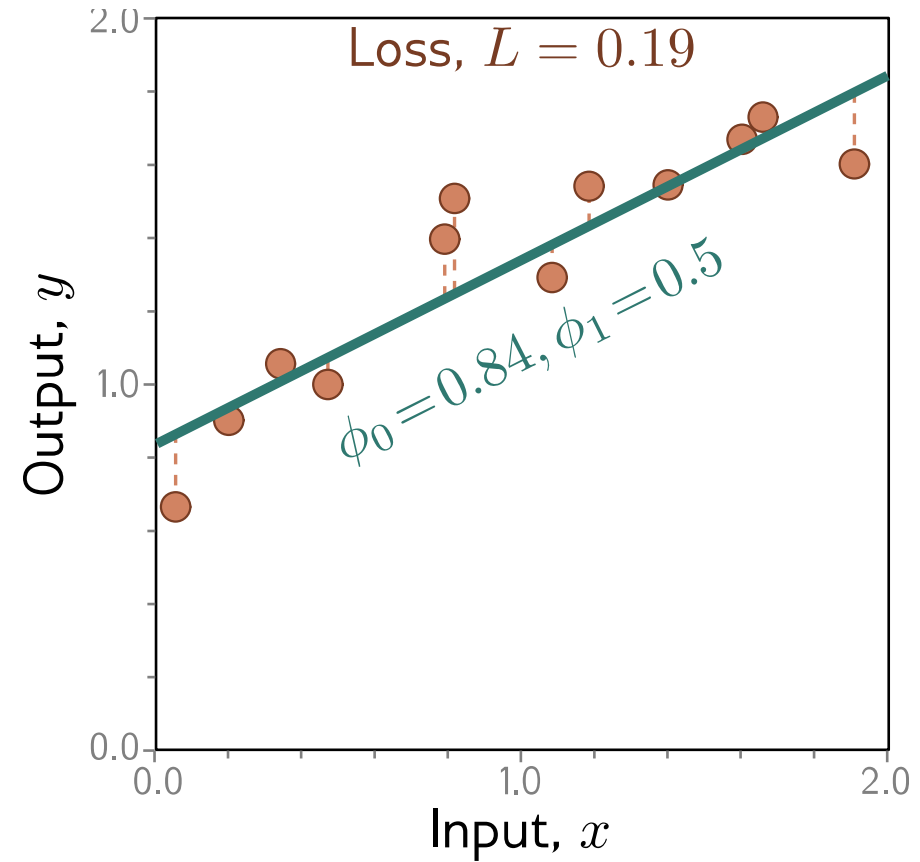
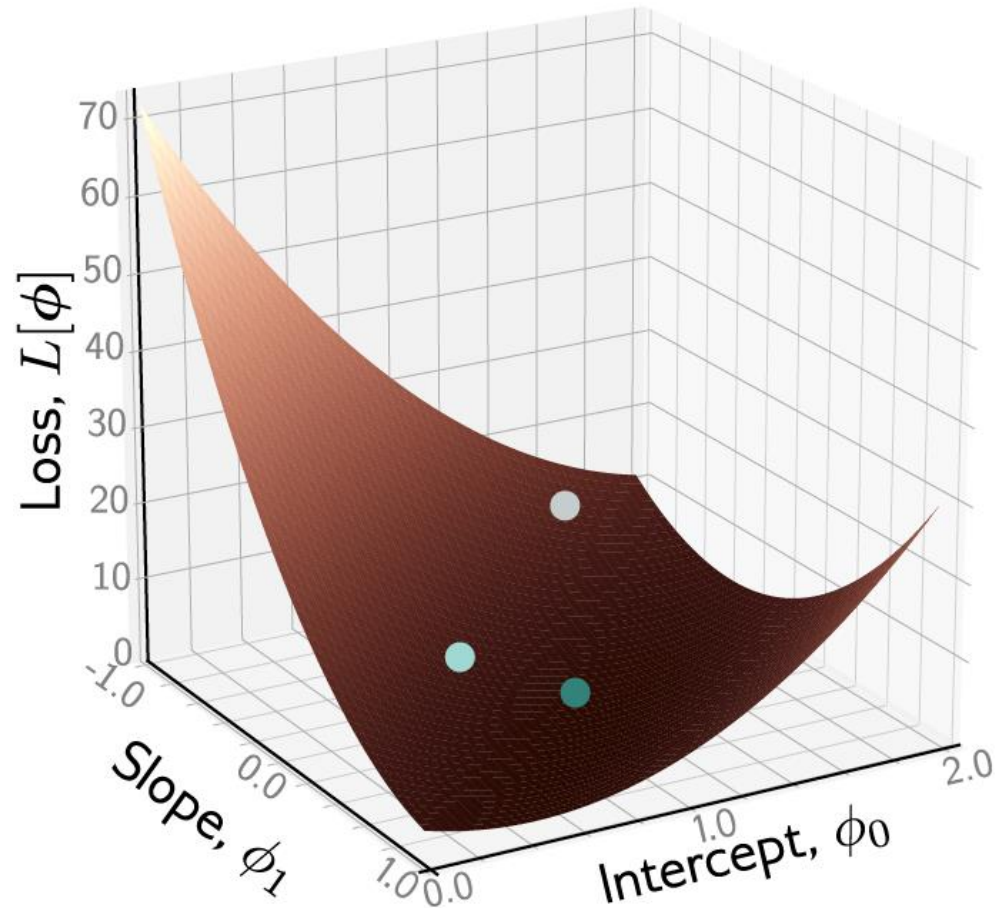
Example: 1D Linear regression loss function



Example: 1D Linear regression loss function

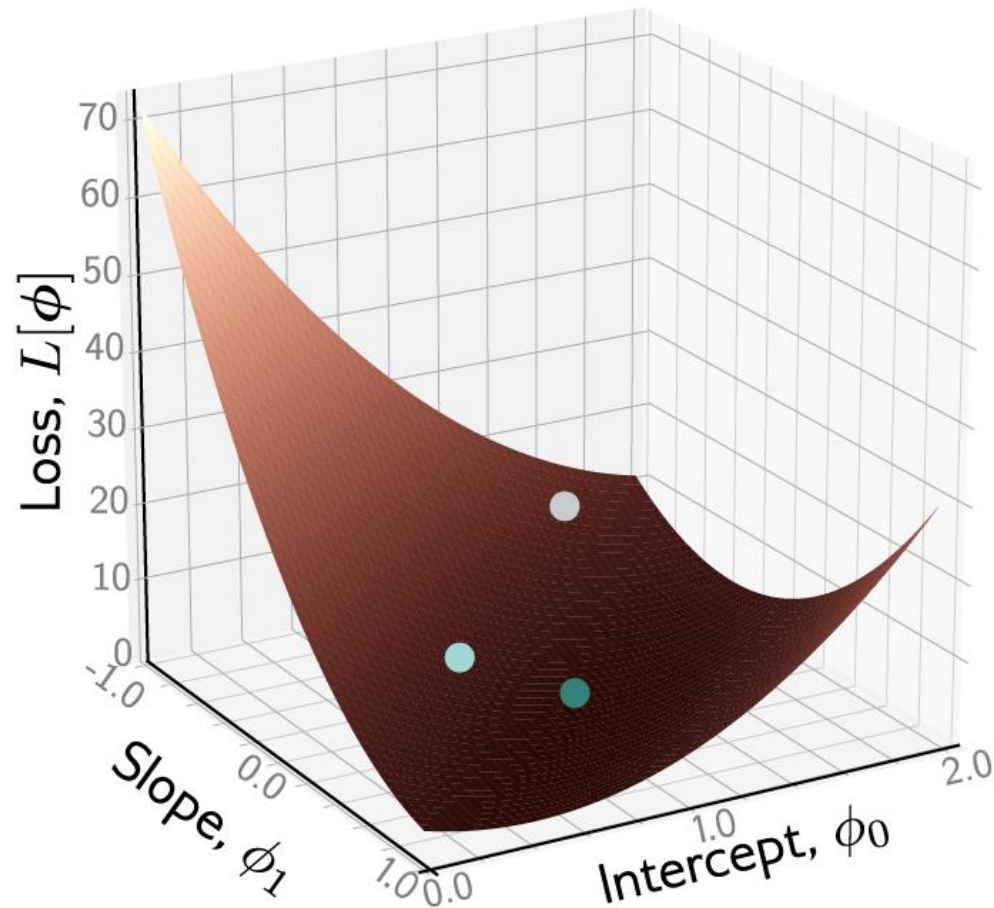


Example: 1D Linear regression loss function

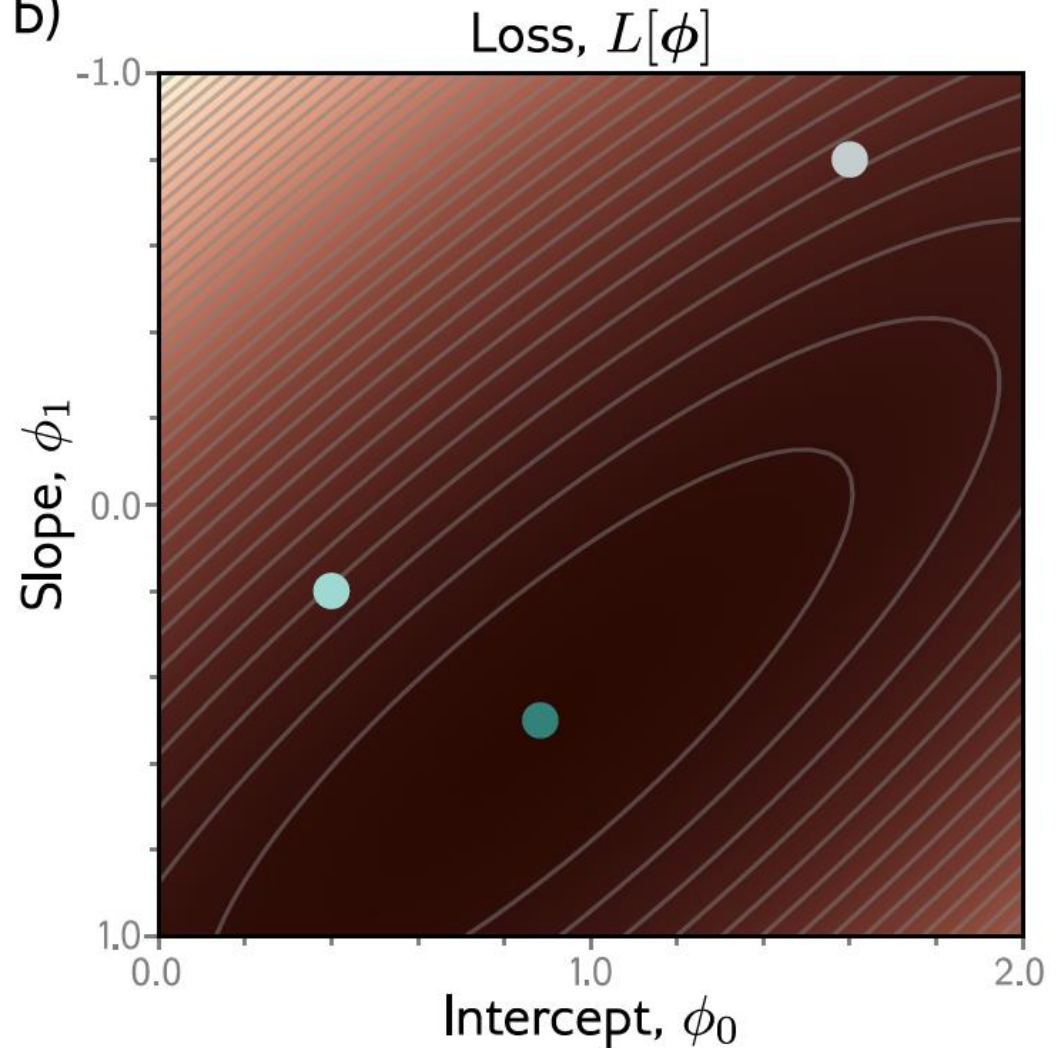


Example: 1D Linear regression loss function

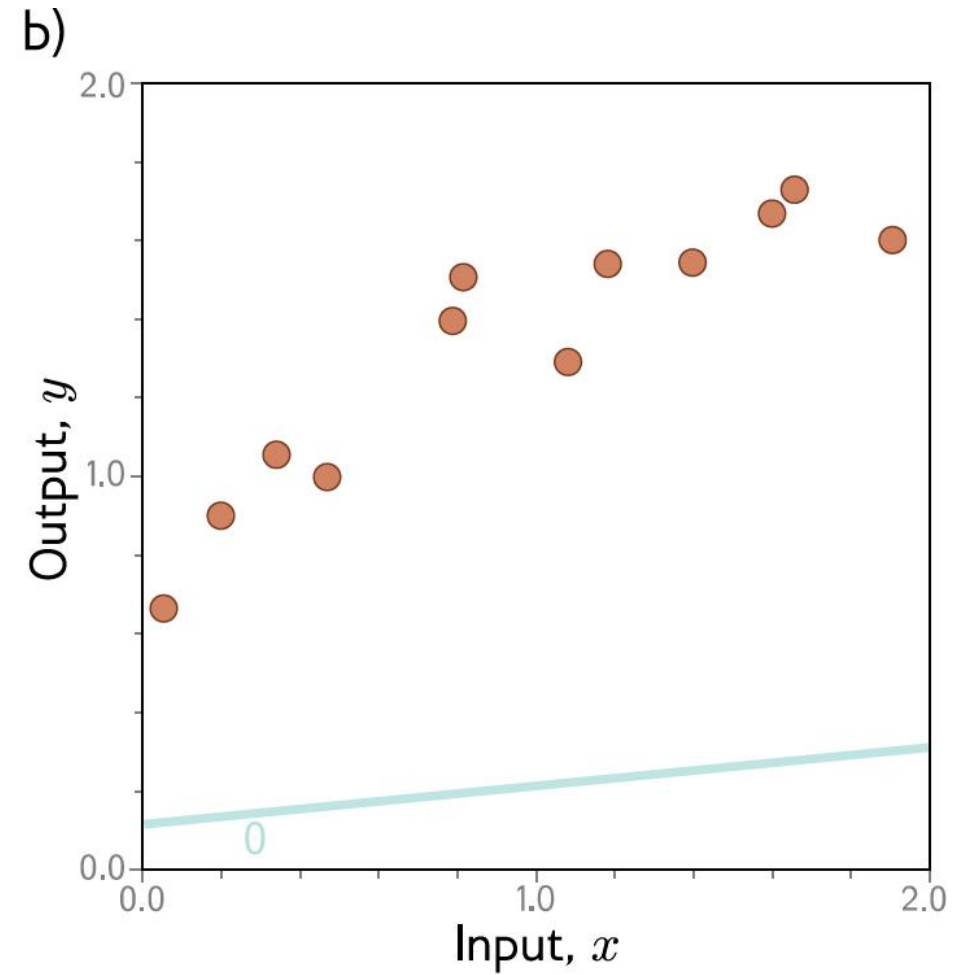
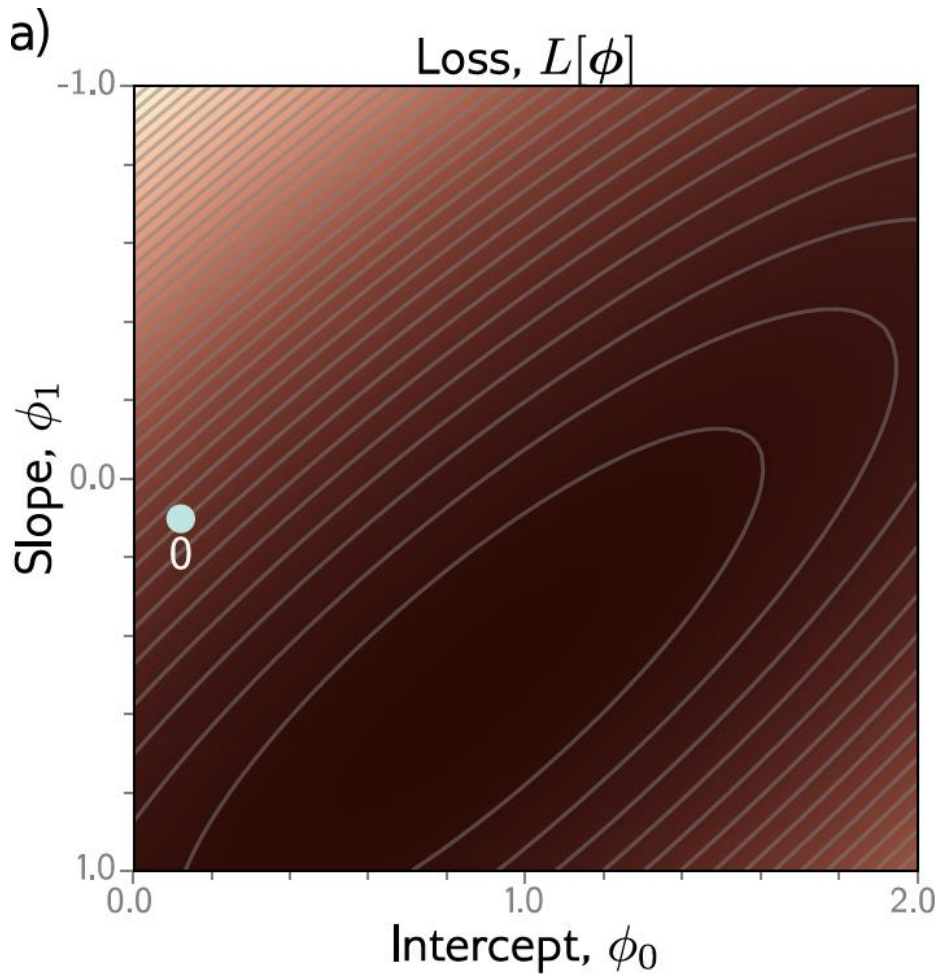
a)



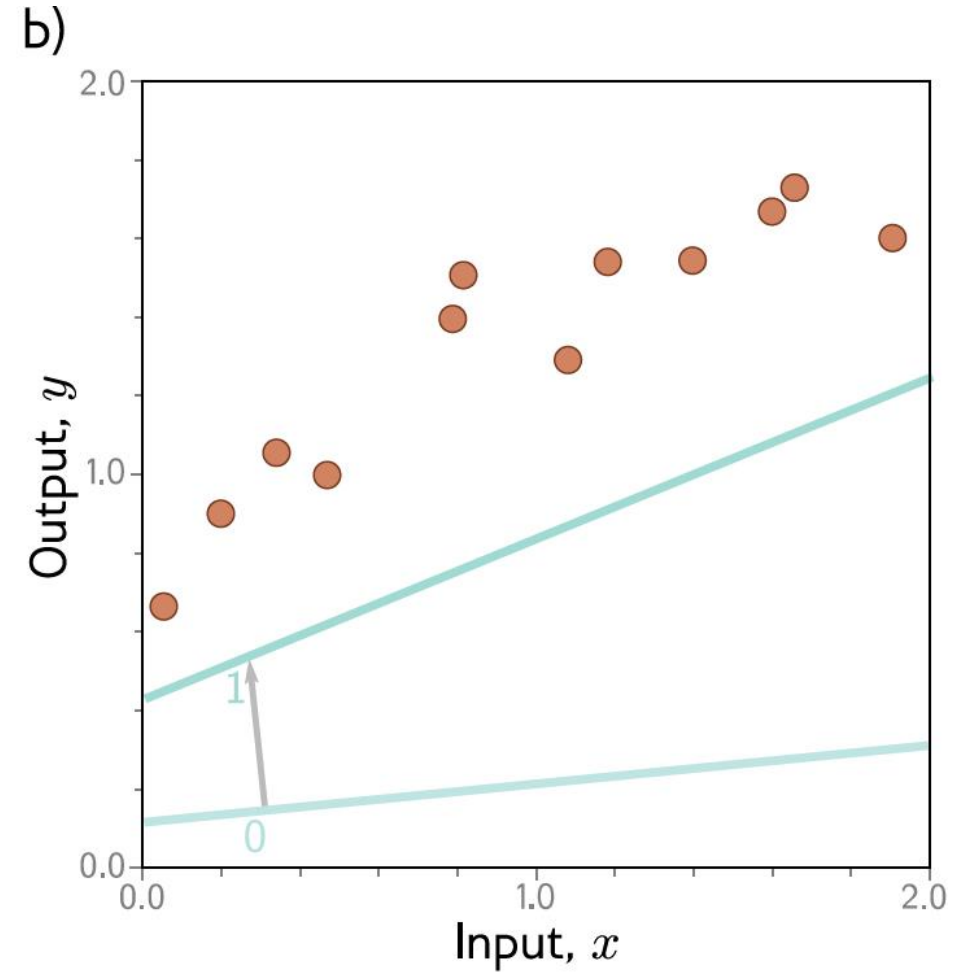
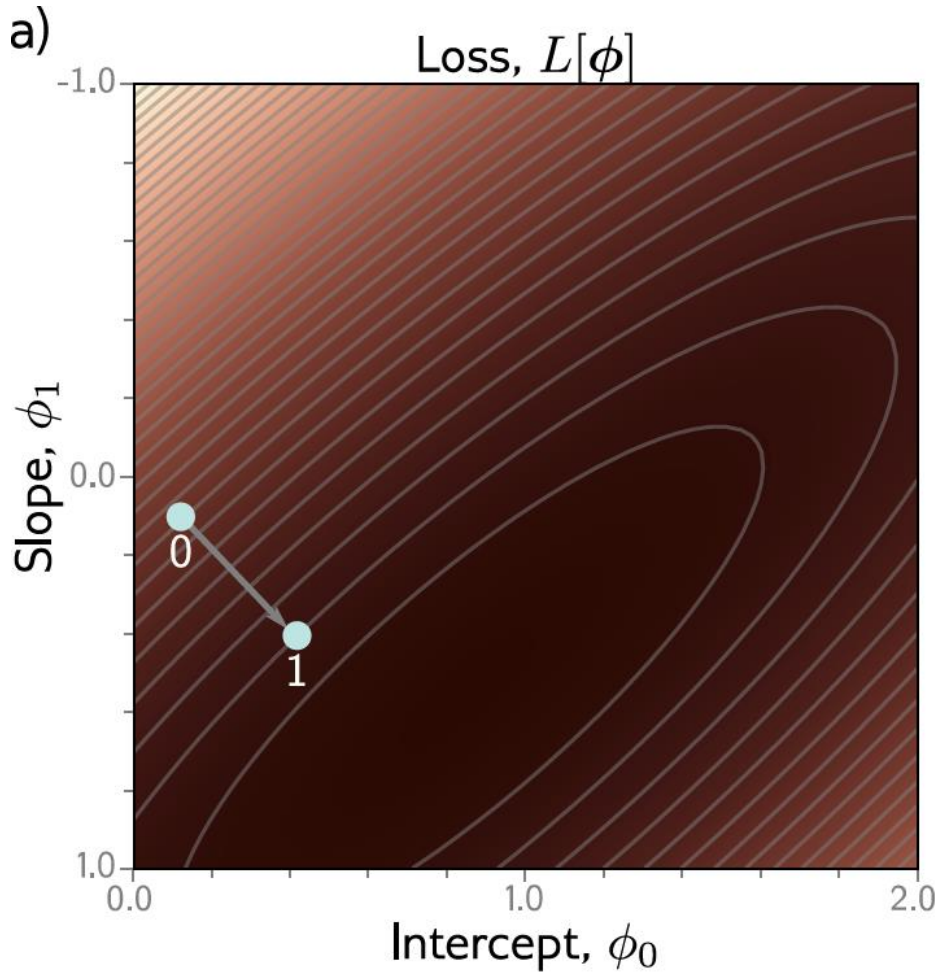
b)



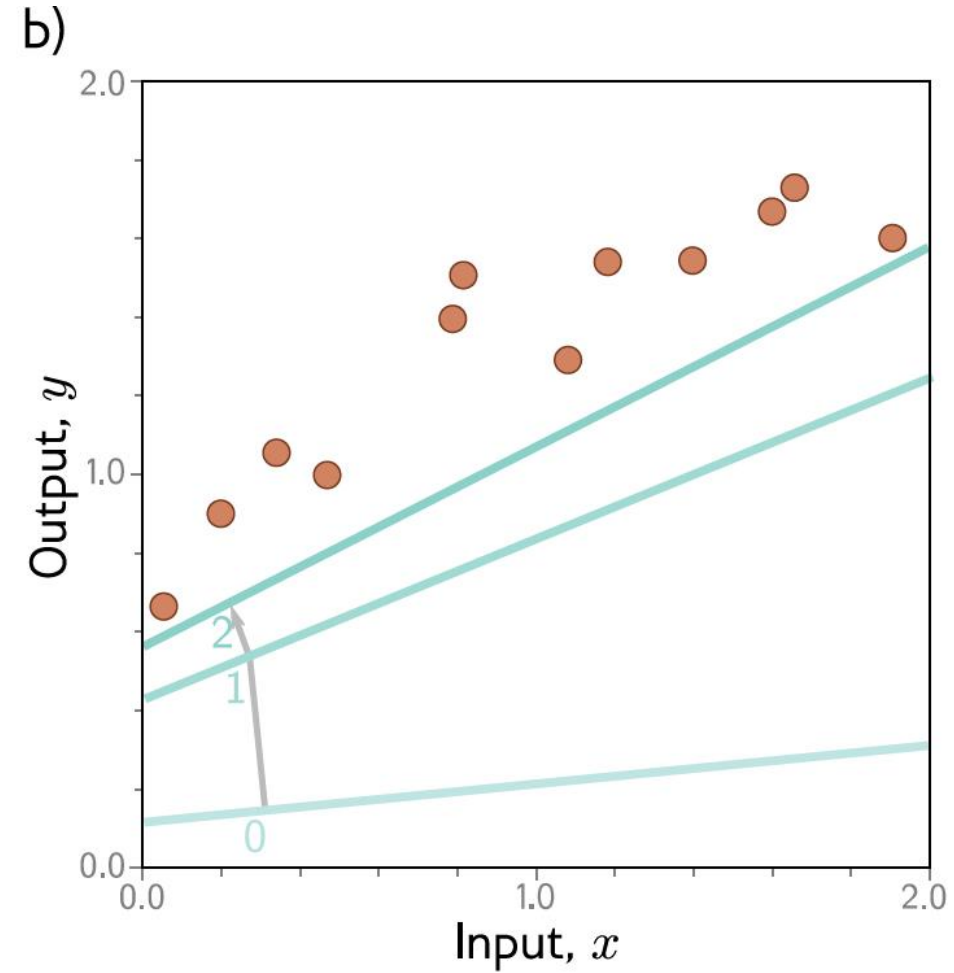
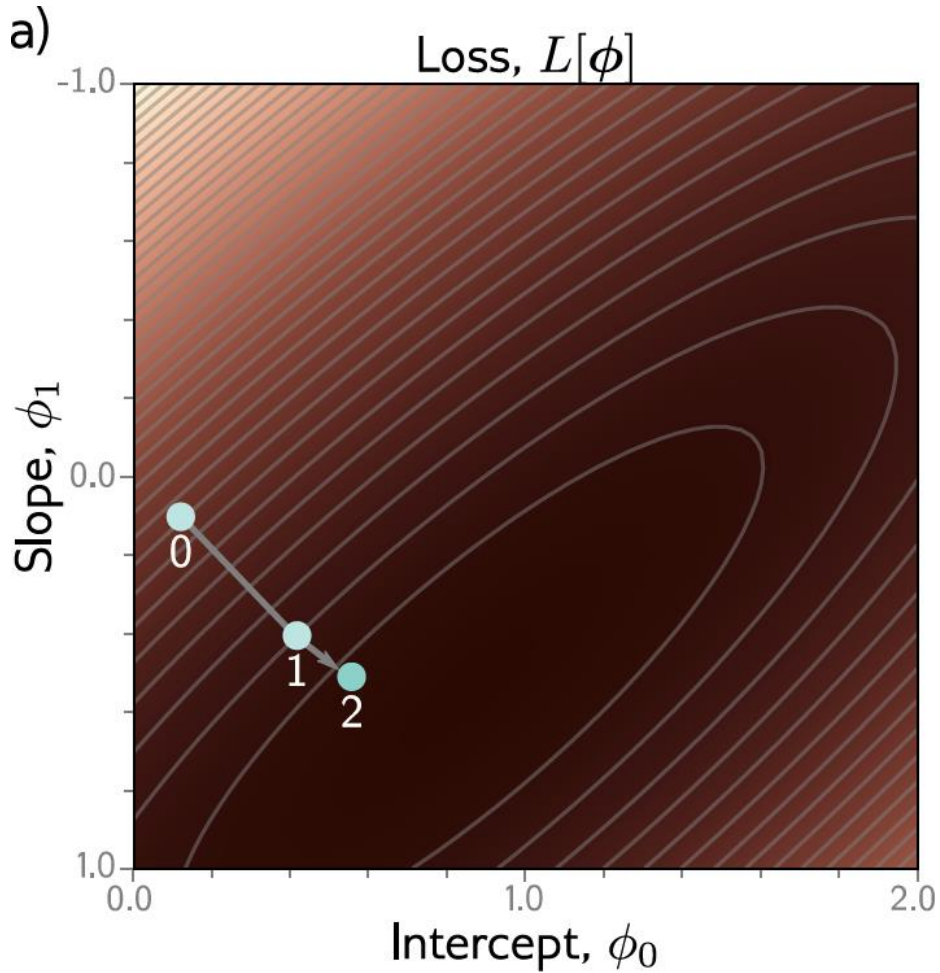
Example: 1D Linear regression training



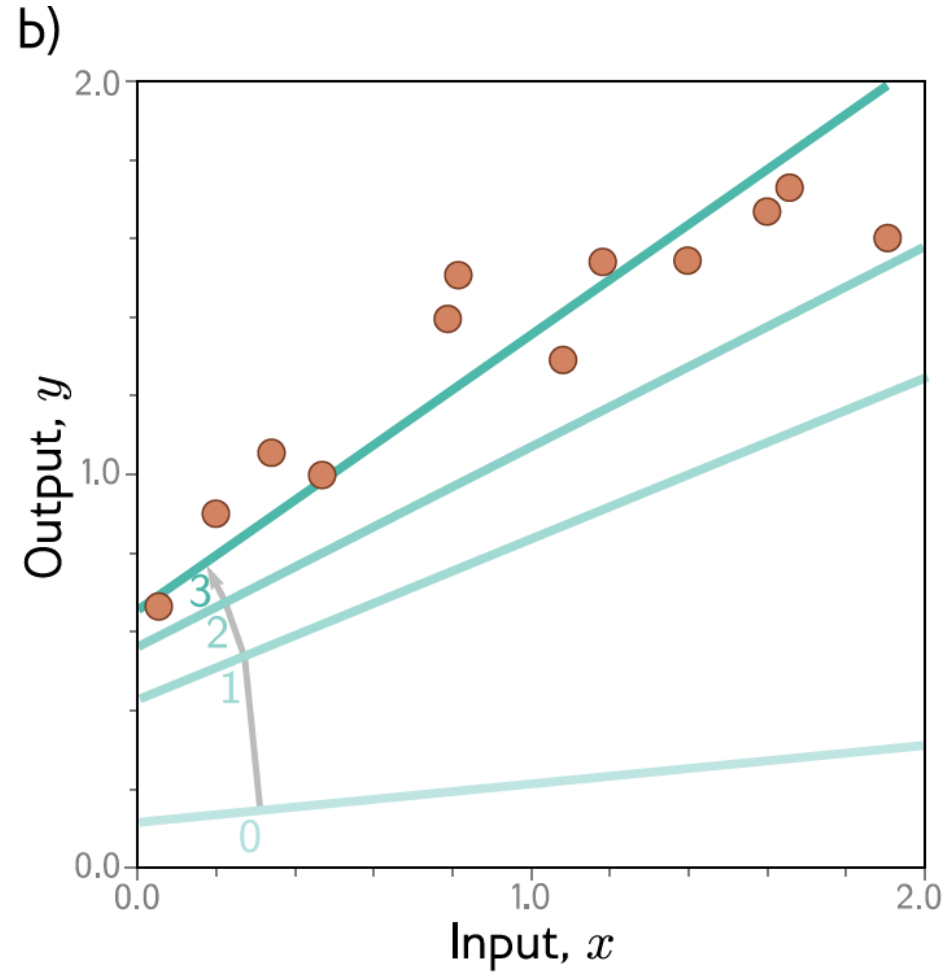
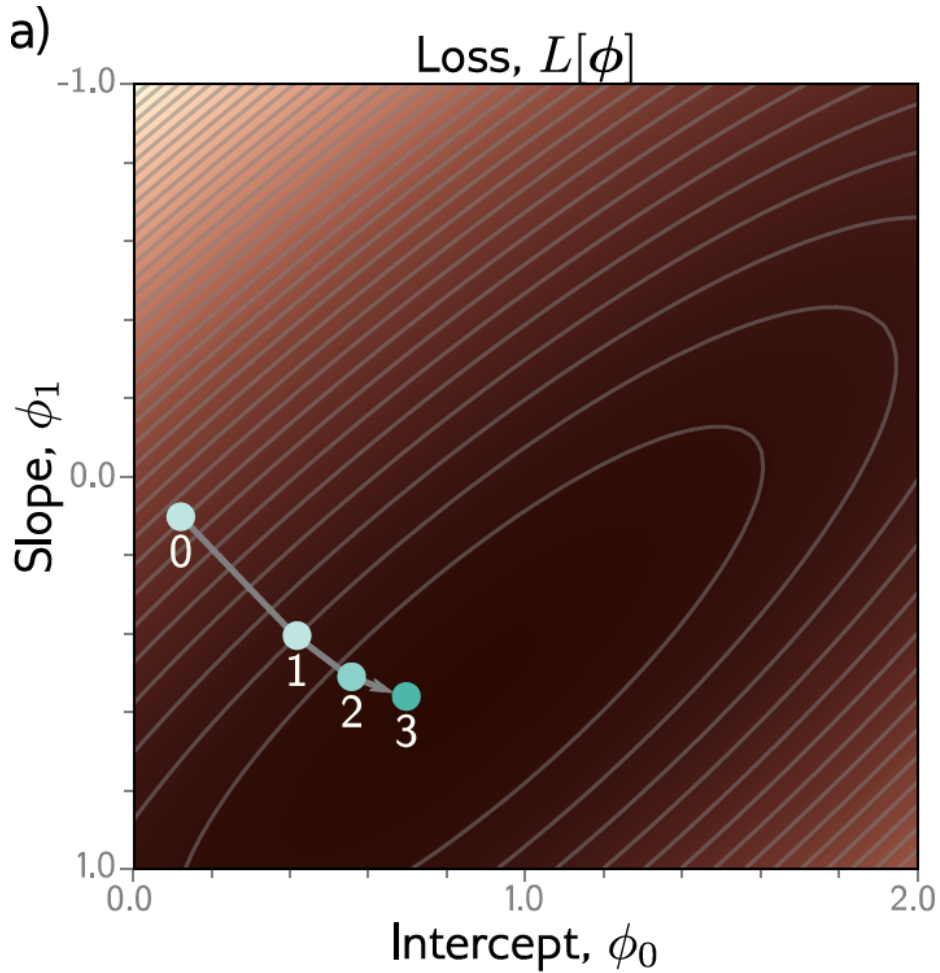
Example: 1D Linear regression training



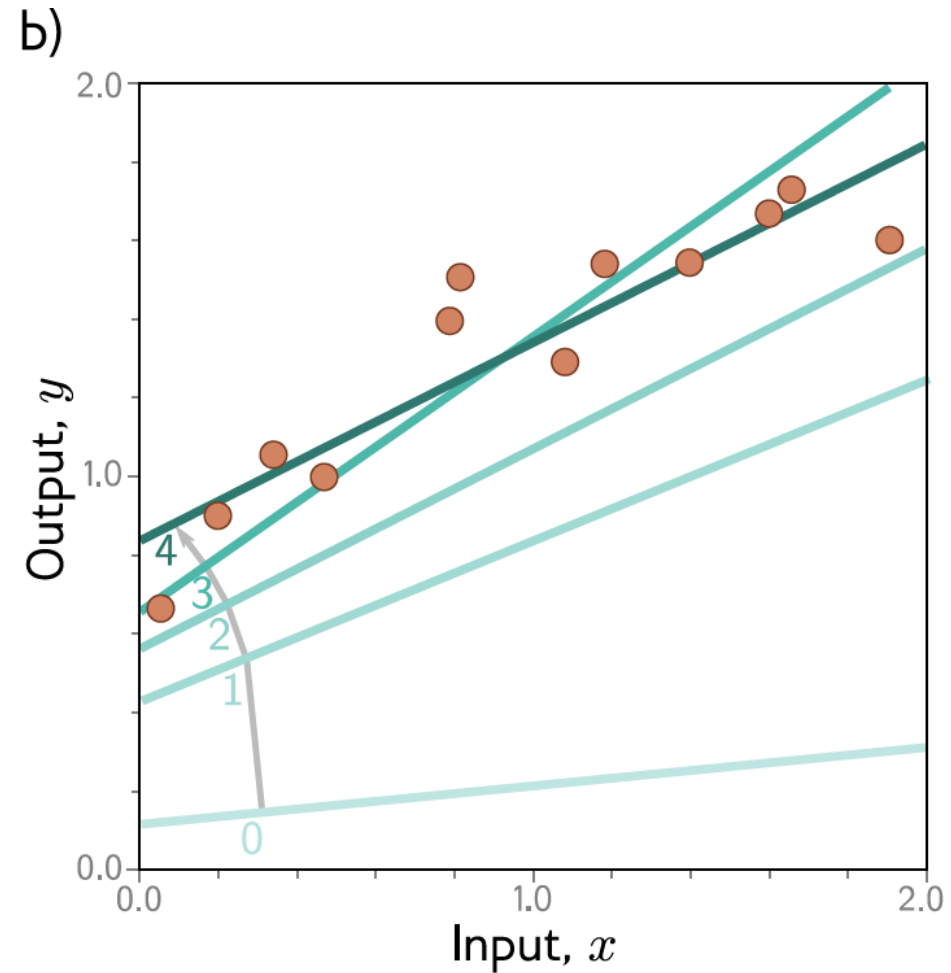
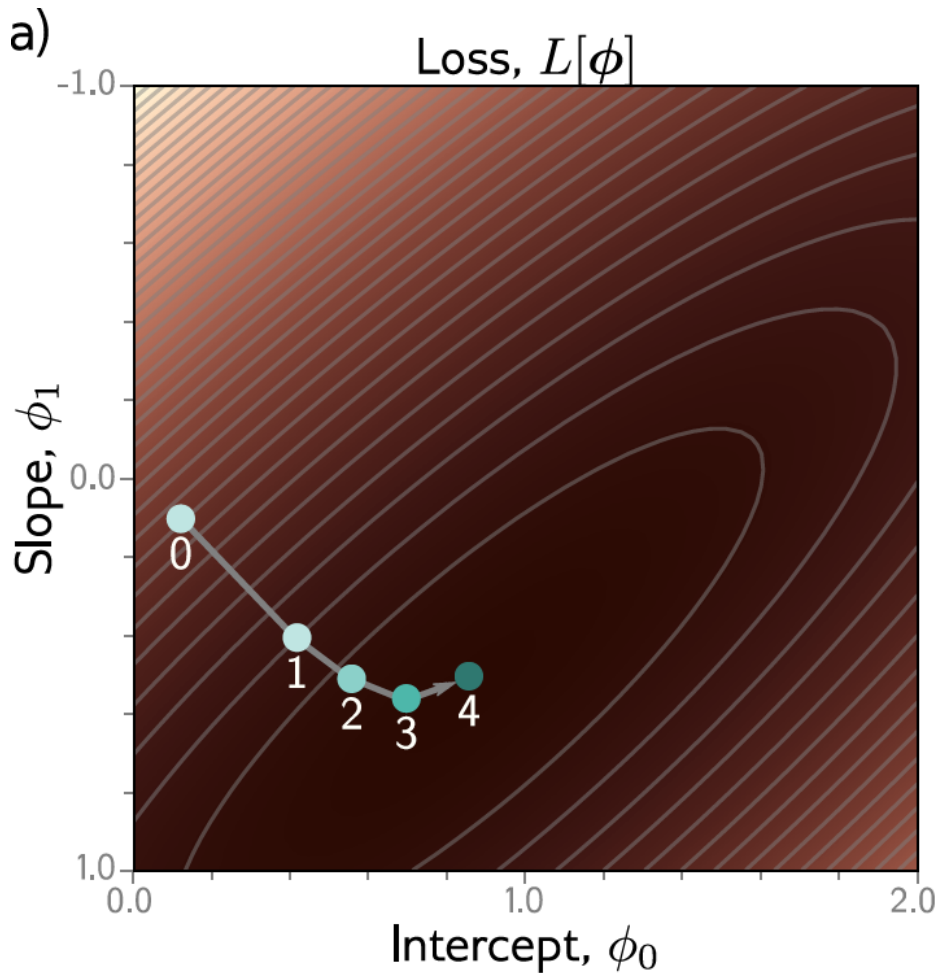
Example: 1D Linear regression training



Example: 1D Linear regression training



Example: 1D Linear regression training

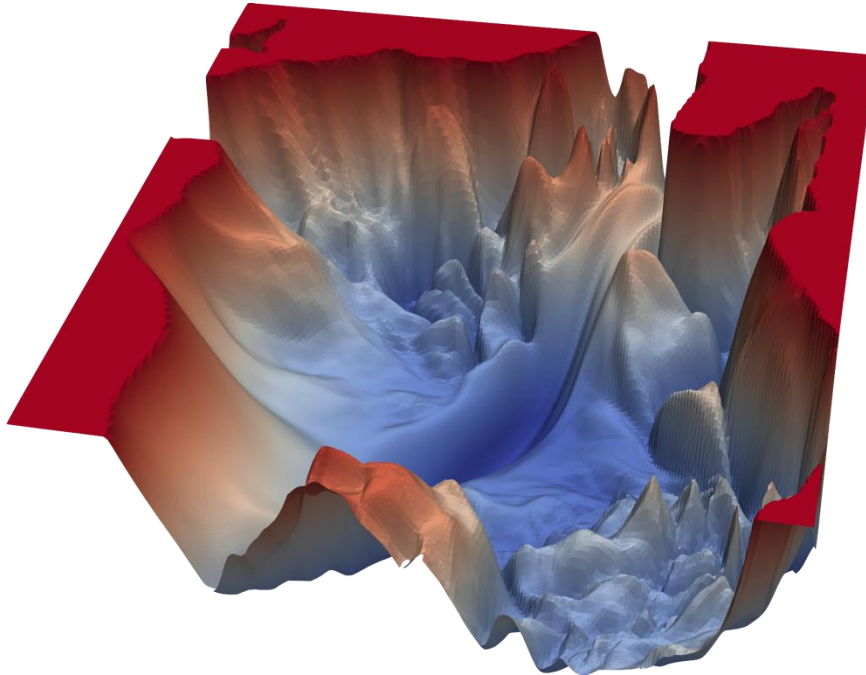


This technique is known as **gradient descent**

[Interactive Figure 2.3](#)

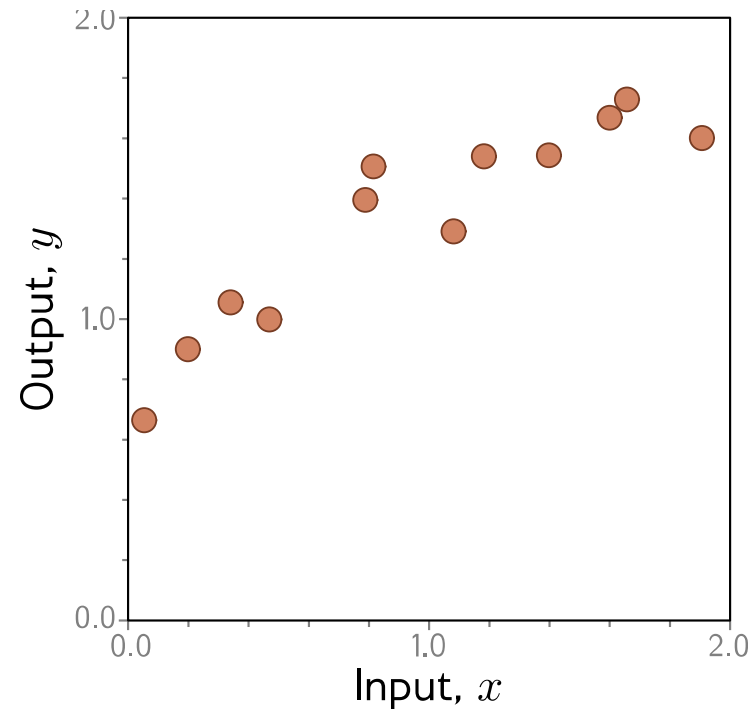
Possible objections

- But you can fit the line model in closed form!
 - Yes – but we won't be able to do this for more complex models
- But we could exhaustively try every slope and intercept combo!
 - Yes – but we won't be able to do this when there are a million parameters



Example: 1D Linear regression testing

- Test with different set of paired input/output data (Test Set)
 - Measure performance
 - Degree to which $Loss$ is same as training = generalization
- Might not generalize well because
 - Model too simple: underfitting
 - Model too complex
 - fits to statistical peculiarities of data
 - this is known as overfitting



Any Questions?

Next Lecture

- How do we choose a loss function in a principled way?